

The Macro Transcension Hypothesis: Spinning Black Holes in Dense Stellar Clusters as Thermodynamic Attractors for Advanced Civilizations, with Omega Centauri as an Observational Test Bed

Tim Swanson

The Omega Centauri Society / Post Oak Labs
tim@postoaklabs.com

September 2026

Draft v2.8 (last revised 2026-09-03) — first paper of the set; companion to the inward-migration review (Paper B), the Omega Centauri campaign (Paper C), the migration economics (Paper D), the engineered-IMBH systems paper (Paper E), the joint accretion bound (Paper F), the residual X-ray census (Paper G), and the mass-tension paper (Paper H); prepared for omegacentauri.me

Abstract

Most proposed resolutions of the Fermi paradox assume that long-lived technological civilizations either expand outward, perish, or deliberately hide. We develop a fourth alternative, the *Macro Transcension Hypothesis* (MTH): that civilizations which optimize for long-term computation are driven by thermodynamics, rather than preference, toward a specific class of astrophysical environment, namely rapidly spinning massive black holes embedded in dense, old stellar systems, and that the migration and its endpoint are both electromagnetically quiet. The MTH extends the transcension hypothesis of [Smart \(2012\)](#) from planet-scale “inner space” to macroscopic black-hole infrastructure, and differs from the aestivation hypothesis of [Sandberg et al. \(2016\)](#) in requiring no waiting strategy: the relevant free-energy and entropy-disposal advantages are available now. We quantify the case in four steps: (i) thin-disk accretion onto a Kerr black hole releases 5.7–30 per cent of rest-mass energy across the spins disk accretion can reach ([Bardeen et al., 1972](#); [Novikov and Thorne, 1973](#); [Thorne, 1974](#)), rising to 42 per cent only in the formal $a_\star \rightarrow 1$ limit that radiative spin equilibrium forbids, versus 0.7 per cent for hydrogen fusion, while magnetically arrested disks extract additional spin energy at effective efficiencies exceeding 100 per cent of accreted rest mass ([Tchekhovskoy et al., 2011](#)); (ii) by the generalized second law ([Bekenstein, 1974](#)), an event horizon is a thermodynamically ideal entropy sink, and a worked delivery budget (Section 4.2) shows the realized erasure cost lands a factor of $\sim 10^6$ – 10^9 below the CMB-limited Landauer floor once carrier propagation and an itemized loss budget are charged, at packet coherence lengths between 0.2 and 3×10^3 km, with little radiated waste heat; (iii) the Bekenstein–Hawking entropy of a $\sim 2 \times 10^4 M_\odot$ black hole corresponds to $\sim 6 \times 10^{85}$ bits, or $\approx 3 \times 10^{85}$ at the Thorne spin the architecture ends at, exceeding any material archive; and (iv) per unit of harvested mass, this architecture outperforms complete fusion of the same fuel by a factor of ~ 11 –60, outperforms Dyson-type stellar harvesting without star lifting by ~ 115 –600 in lifetime energy yield (~ 11 –60 against a star-lifting economy), and exceeds a solar Dyson swarm by a factor of $\sim 10^9$ (6.6×10^8 at $2 \times 10^4 M_\odot$) in instantaneous Eddington-limited power.

We then identify Omega Centauri (NGC 5139), a $\sim 4 \times 10^6 M_\odot$, ~ 12 -Gyr-old stripped dwarf-galaxy nucleus hosting the nearest strong candidate intermediate-mass black hole ([Häberle et al., 2024b](#)),

as the most observationally accessible system satisfying the MTH selection criteria, and present a falsification framework built on six instrument-matched tests: accretion-luminosity limits (JWST, ATCA), mid-infrared waste-heat limits (JWST), a gravitational-wave signature inconsistent with a single point mass, an EMRI point-mass measurement, and an EMRI spin measurement (the latter three via LISA, contingent on a compact-object inspiral occurring in-band during the mission), plus neutrino burst searches (KM3NeT). Current data, including the unresolved tension between a $\geq 8,200 M_{\odot}$ kinematic lower bound (Häberle et al., 2024b) and a $\lesssim 6,000 M_{\odot}$ pulsar-timing upper bound (Bañares-Hernández et al., 2025) and the complete electromagnetic silence of the central object (Mahida et al., 2026; Chen et al., 2026), are consistent with both the MTH and the more parsimonious gas-starvation null hypothesis; we state explicitly which forthcoming observations would discriminate between them, and which would falsify the MTH outright. The hypothesis is offered in the falsificationist tradition of Sandberg et al. (2016) and Dvali and Osmanov (2023): a speculative but physically grounded working model whose value lies in the concrete observational program it motivates.

Keywords: Fermi paradox; technosignatures; SETI; intermediate-mass black holes; Omega Centauri; NGC 5139; Blandford–Znajek mechanism; Landauer limit; transcension; aestivation

Contents

1	Introduction	4
1.1	Claims	5
1.2	Structure of the paper	5
2	Relation to prior work	6
2.1	The expansionist mainstream	6
2.2	The inward-migration family	6
2.3	New content of the hypothesis	8
3	The hypothesis	8
3.1	Selection criteria over host systems	9
4	The thermodynamic and computational case	11
4.1	Energy: accretion and spin extraction versus fusion	11
4.2	Entropy disposal: the horizon as the ultimate cold reservoir	14
4.3	Information storage: the Bekenstein–Hawking archive	17
4.4	Time dilation as an engineering parameter	18
4.5	Inward versus outward migration	19
4.6	One hole rather than many	19
5	Omega Centauri as the optimal observational target	20
5.1	The cluster	20
5.2	The enclosed-mass profile	21
5.3	The IMBH candidate and the mass tension	22
5.4	Electromagnetic silence	23
5.5	The case against Sagittarius A*	24
6	Falsification framework and observational program	24
6.1	Supporting evidence criteria	27
7	Discussion: objections and limits	28
8	Conclusion	29
A	Staged exploitation architecture	40

1 Introduction

The Fermi paradox, in its modern form (Hart, 1975; Tipler, 1980; Webb, 2015; Ćirković, 2018; Forgan, 2019), rests on an expansionist premise: that at least some technological civilizations should grow outward across interstellar distances, and that such growth, sustained for even a small fraction of galactic history, would be conspicuous. The premise is quantitatively robust. Self-reproducing probe architectures permit galaxy-scale colonization in 10^6 – 10^8 years (Tipler, 1980; Freitas and Gilbreath, 1982), intergalactic expansion is energetically cheap for a mature civilization (Armstrong and Sandberg, 2013), and surveys for the waste heat of large-scale energy harvesting have returned null results across $\sim 10^5$ galaxies (Wright et al., 2014; Griffith et al., 2015). The observed silence therefore demands either that technological life is extremely rare, that it reliably destroys itself, or that the expansionist premise is wrong.

A minority tradition in the literature attacks the premise itself. Smart (2012) proposed the *transcension hypothesis*: that the developmental trajectory of intelligence points not outward but inward, toward ever denser, faster, and more efficient configurations of matter and energy (“STEM compression” of space, time, energy, and matter), terminating, in Smart’s telling, at black-hole-scale densities and an effective exit from the observable universe. Sandberg et al. (2016) proposed the *aestivation hypothesis*: that civilizations which value computation should defer it to the far future, when the cosmic microwave background (CMB) has cooled and each erased bit costs less free energy, and should therefore be dormant now. Both proposals share a key insight, that the thermodynamics of computation is the correct lens through which to predict the behaviour of mature intelligence, and both have been criticized on physical grounds. Smart’s mechanism for leaving the universe is not specified in testable form, and the aestivation argument was shown by Bennett et al. (2019) to rest on a thermodynamic error: free energy not harvested now is largely lost, not banked, so a computation-maximizing civilization should collect resources now even if it spends them later.

This paper develops a third member of that family, which we call the *Macro Transcension Hypothesis* (MTH). Its core claim is that the inward trajectory identified by Smart has a concrete, macroscopic, observationally accessible destination in ordinary astrophysics: rapidly spinning massive black holes embedded in dense, dynamically old stellar systems. No new physics, no exit from the universe, and no multi-gigayear dormancy are required. The advantages that drive the migration (accretion power at up to ~ 42 times the mass-energy efficiency of hydrogen fusion at the highest spin disk accretion reaches, an event horizon serving as an entropy sink of unlimited capacity, information storage at the Bekenstein–Hawking density, and gravitational time dilation as a controllable computational resource) are all available in the present epoch, and all follow from textbook general relativity and black-hole thermodynamics (Misner et al., 1973; Bardeen et al., 1972; Bekenstein, 1973, 1974). The hypothesis predicts that the most advanced civilizations remain present but electromagnetically invisible, because maximum computational efficiency, viewed from outside, is electromagnetically quiet.

The MTH would remain idle philosophy if it did not select targets. The selection criteria (a massive black hole of modest depth, intermediate rather than supermassive; a dense retinue of low-mass stars as fuel; dynamical age sufficient for any migrating civilization to have arrived and matured; and a quiescent electromagnetic environment) converge on a short list of Galactic systems, at the top of which sits Omega Centauri (ω Cen, NGC 5139): the Milky Way’s most massive globular cluster, widely interpreted as the stripped nucleus of an accreted dwarf galaxy (Hilker and Richtler, 2000; Bekki and Tsujimoto, 2019; Ibata et al., 2019), host to $\sim 10^7$ stars of mean age 12.08 ± 0.01 Gyr (Clontz et al., 2024), and, since the detection of seven fast-moving stars within its central 3 arcsec (five of which are robust detections), host to the strongest current candidate for an intermediate-mass black hole (IMBH) in the Galaxy (Häberle et al., 2024b). That candidate is simultaneously the subject of a sharp observational controversy: combined stellar-kinematic and millisecond-pulsar-timing analysis places a 3σ upper limit of $\sim 6,000 M_\odot$ on any

central point mass (Bañares-Hernández et al., 2025), formally inconsistent with the kinematic lower bound. And it is electromagnetically dark to remarkable depth: neither ~ 170 hours of ATCA radio integration (Mahida et al., 2026), nor JWST infrared photometry (Chen et al., 2026), nor a ~ 291 ks Chandra exposure (Haggard et al., 2013) detects any accretion signature. Every element of this situation (a contested central mass, perfect electromagnetic silence, an imminent decisive instrument in LISA) makes ω Cen a natural laboratory for testing inward-migration hypotheses, entirely independent of whether the MTH is true.

1.1 Claims

Because the argument mixes established physics with speculative inference, we state the epistemic register of each claim explicitly. The paper makes three claims of different strength:

1. **(Established physics plus a comparative synthesis.)** Spinning black holes in dense old clusters are, by a wide quantitative margin, the most thermodynamically favourable environments in the present-day universe for long-term, large-scale computation. The ingredient results are established physics; the optimality claim is a synthesis over a stated comparison set (fusion, Dyson-type stellar harvesting, and cold-space computing), defended quantitatively in Section 4 using standard results (Novikov and Thorne, 1973; Blandford and Znajek, 1977; Bekenstein, 1974; Landauer, 1961; Lloyd, 2000), and is independent of any claim about extraterrestrial intelligence. The ranking of cluster IMBHs above supermassive holes rests partly on strategic rather than thermodynamic criteria (hazard exposure, contested resources), which we treat separately in Section 5.5.
2. **(Speculative hypothesis.)** If long-lived technological civilizations exist and even a subset optimizes for computational capacity (Sandberg, 1999), the environments of claim 1 act as attractors, and the resulting migration resolves the Fermi paradox without rare-Earth or doomsday assumptions. This is the MTH proper, stated formally in Section 3. We regard it as less parsimonious than the null hypothesis (technological life is rare or absent) and say so.
3. **(Observational program.)** Whether or not claim 2 is true, it generates instrument-matched, time-bounded, falsifiable predictions at ω Cen, several of which will be adjudicated by existing or funded instruments within ~ 15 years. This program is presented in Section 6.

We do not claim that ω Cen hosts a civilization; we do not claim the IMBH is confirmed (it is not); and we do not claim the observed electromagnetic silence is evidence for the MTH, since it is equally consistent with the simpler gas-starvation explanation that we adopt as the null hypothesis throughout.

1.2 Structure of the paper

Section 2 situates the MTH in the literature. Section 3 states the hypothesis formally as four postulates. Section 4 develops the quantitative thermodynamic and computational case. Section 5 reviews ω Cen and its IMBH evidence, including the current mass tension. Section 6 presents the falsification framework and observational roadmap; the staged architecture that motivates it, sketched at the level of explicitly labelled speculation, is developed in Appendix A. Section 7 addresses objections, and Section 8 concludes. Figure 1 previews the argument chain and its epistemic register.

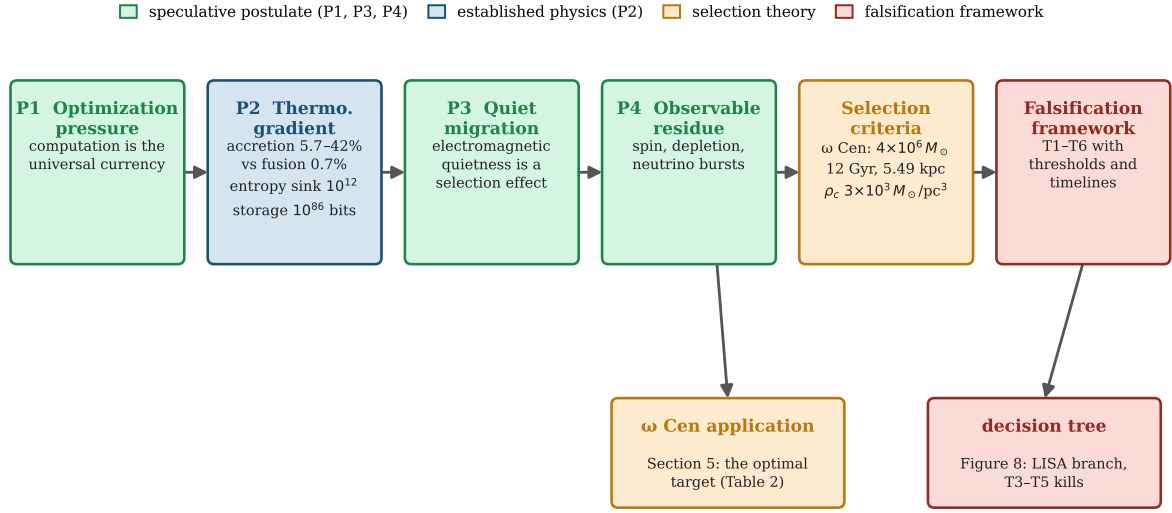


Figure 1: The argument chain of the paper as a left-to-right cascade, coloured by the epistemic register of Section 1.1: green, the speculative postulates (P1, P3, P4); blue, the established physics of P2 (Section 4); orange, the selection theory (Section 3.1); red, the falsification framework (Section 6, Table 7). Each box carries the quantitative anchor developed in the section it points to. The observable residue P4 branches to the target application (Section 5), and the framework terminates in the decision tree of Figure 7.

2 Relation to prior work

2.1 The expansionist mainstream

The canonical statements of the Fermi paradox (Hart, 1975; Tipler, 1980) treat interstellar expansion as the default behaviour of technological life, an assumption sharpened by Armstrong and Sandberg (2013), who showed that a single mature civilization could seed every galaxy within several billion light-years using a fraction of one star’s output. The “grabby aliens” model of Hanson et al. (2021) quantifies the selection effect: if loud (expanding) civilizations exist at any appreciable rate, they dominate the future light cone, and our own early arrival is informative about their density. Observational programs built on the expansionist premise (radio and optical SETI, Tarter 2001; waste-heat surveys, Dyson 1960; Wright et al. 2014; Griffith et al. 2015; and the broader technosignature framework, Wright et al. 2022; Socas-Navarro et al. 2021; Lingam and Loeb 2021) have produced consistent null results. Within the expansionist frame, these nulls force the unattractive trilemma of rarity, doom, or concealment.

2.2 The inward-migration family

The MTH belongs to a smaller family of proposals holding that mature intelligence migrates down thermodynamic gradients rather than out across space.

Transcension. Smart (2012) argued that the developmental history of complexity on Earth exhibits sustained compression in space, time, energy, and matter (“STEM compression”), and extrapolated to a future in which civilizations engineer environments of black-hole density and disappear from the observable universe. The MTH adopts the compression premise but rejects the terminal step: nothing in known physics permits or requires an exit from the universe, and a hypothesis terminating in unobservable territory cannot be tested. The MTH instead halts the inward trajectory at a macroscopic astrophysical

object, an existing massive black hole, where every claimed advantage is calculable and several are observable, which is the sense in which the transcension here is macro.

Postbiological migration. The closest precursor to the present proposal is [Ćirković and Bradbury \(2006\)](#), who argued that postbiological civilizations should migrate down Galactic thermodynamic gradients toward the cold outer Galaxy, where computation is cheaper, and that SETI’s apparent failure reflects a search aimed at the wrong locations. The MTH shares the thermodynamic-migration logic and the diagnosis of SETI’s targeting problem, but identifies a different optimum: the relevant gradient, we argue, is set by the conjunction of energy density, entropy-disposal capacity, and storage density rather than by ambient temperature alone, and its maxima are spinning black holes in dense old clusters rather than the Galactic outskirts.

Aestivation. [Sandberg et al. \(2016\)](#) proposed that civilizations defer computation until the CMB cools, gaining a factor of up to $\sim 10^{30}$ in computations per joule, and so should currently be dormant. [Bennett et al. \(2019\)](#) showed the argument neglects the irreversible loss of free-energy resources not collected promptly: stars burn whether or not anyone harvests them, so maximizers should be active now. The MTH avoids this objection by construction: instead of waiting for a colder universe, it imports a colder-than-CMB entropy sink into the present. Waste entropy crossing an event horizon increases horizon area in accordance with the generalized second law ([Bekenstein, 1974](#)), at an effective marginal temperature far below the CMB’s 2.7 K (Section 4.2). An MTH civilization harvests aggressively in the present epoch, consistent with the Bennett–Hanson–Riedel optimality argument, while enjoying much of the efficiency gain that motivated aestivation.

Black holes as civilization infrastructure. The conceit is older than SETI’s engagement with it. [Penrose \(1969\)](#) introduced rotational-energy extraction with an advanced-civilization framing, and [Misner et al. \(1973, §33.7\)](#) formalized it as the garbage-dump thought experiment: a civilization lowers its waste to just outside the horizon, releases it, and recovers up to 29 per cent of the rest mass as work. [Thorne \(1994\)](#) staged the same power station for a popular audience, and [Frolov and Novikov \(1998, §12\)](#) give the textbook treatment of black holes as energy sources for civilizations. The present series is a completion of that thread. Within SETI proper, several authors have treated black holes as energy sources or computational substrates for advanced intelligence: [Inoue and Yokoo \(2011\)](#) proposed Dyson-type collectors around an accreting supermassive black hole; [Hsiao et al. \(2021\)](#) computed the detectability of a “Dyson sphere around a black hole” harvesting disk luminosity; [Opatrný et al. \(2017\)](#) analysed the inverted configuration in which a black hole serves as the cold reservoir of a heat engine run against the warmer sky; [Vidal \(2011, 2014\)](#) argued that black-hole-coupled (“stellivore”) systems are the natural attractor for cosmic intelligence; and [Dvali and Osmanov \(2023\)](#) analysed black holes as quantum information processors for extraterrestrial civilizations, predicting characteristic high-energy neutrino emission from artificial micro-black-hole computers. [Lacki and DiKerby \(2025\)](#) survey the corresponding high-energy SETI opportunity, and the Dyson Minds workshop ([Curtis et al., 2026](#)) has since brought this thread into mainstream observational planning, recommending anomaly-detection searches of archival WISE, JWST, and EHT data for black-hole-hosted intelligence, with a focus on supermassive hosts. The MTH synthesizes this thread and adds the two elements it has lacked: a *selection theory* specifying which black holes are preferred and why (intermediate-mass, high-spin, cluster-embedded; Section 3.1), and a named, ranked observational target with a costed test program.

Kardashev and his inversion. The Kardashev scale ([Kardashev, 1964](#)) ranks civilizations by total energy throughput, implicitly rewarding outward expansion. [Barrow \(1998\)](#) proposed the complementary inward scale: mastery over ever-smaller scales of structure. The MTH is the astrophysical completion of Barrow’s inversion — the observation that the universe already provides, free of charge, the densest stable structures that inward mastery could aim at, and that the rational endpoint of a Barrow-type trajectory

is therefore relocation rather than fabrication. Bradbury (1999) reached a related conclusion from the engineering side: his Matrioshka-brain analysis shows stellar-powered computation is ultimately limited by radiator area and sink temperature, the constraints an event horizon eliminates.

2.3 New content of the hypothesis

To our knowledge this paper is the first to (i) state the inward-migration resolution of the Fermi paradox as a falsifiable selection theory over real astrophysical objects; (ii) integrate the generalized second law, the Landauer bound, Kerr accretion efficiency, Blandford–Znajek extraction, and Bekenstein–Hawking storage into a single quantitative budget for a civilization-scale computer hosted by an existing black hole; and (iii) attach the resulting predictions to a specific system (ω Cen) with instrument-matched falsification thresholds and timelines. Each ingredient exists in the literature; the synthesis, the selection criteria, and the test program are new.

3 The hypothesis

Epistemic status: Sections 3–4 contain, respectively, a speculative hypothesis stated formally, and established physics offered in its support. The physics in Section 4 stands regardless of the hypothesis.

We state the MTH as four postulates.

P1 (Optimization pressure). Among long-lived technological civilizations, a non-empty subset evolves (by competition, self-modification, or design) toward maximizing long-term computational capacity per unit of controlled resource, under at least one of three conditions on the objective: (i) temporal discounting, (ii) a requirement that the computation be unified and latency-bound, or (iii) a bounded planning horizon. The three are needed because an undiscounted maximizer of total integrated computation over unbounded time is served by expansion, for reasons set out in Section 4.5; they are stated here rather than added later. (Motivation: computation is the universal currency into which information-processing agents can convert essentially any terminal goal. This is the instrumental-convergence thesis (Omohundro, 2008; Bostrom, 2012), given formal treatment by Benson-Tilsen and Soares (2016), and the same premise adopted by Smart 2012 and Sandberg et al. 2016. What remains open is the step from convergent instrumental goals in a single agent to a population claim over lineages; P1 asserts only a non-empty subset, the weakest form that step permits.)

P2 (Thermodynamic gradient). For any such maximizer, the present-day universe contains a strict gradient of substrate quality whose global optima are rapidly spinning massive black holes embedded in dense, old, low-luminosity stellar systems. The gradient is set by four independent physical quantities: extractable energy per unit fuel mass, entropy-disposal capacity, information-storage density, and control over effective clock rate. (Defended quantitatively in Section 4.)

P3 (Quiet migration). Migration toward and exploitation of such environments is electromagnetically inconspicuous at interstellar distances. Transit uses gram-to-kilogram-scale beamed-sail payloads rather than worldships (Lubin, 2016); construction uses in-situ resources (Freitas and Gilbreath, 1982); and mature operation, by the very efficiency that motivates it (P2), radiates negligible waste heat (Section 4.2).

P4 (Observable residue). The endpoint is nevertheless not perfectly unobservable. It predicts (a) anomalously high spin on the host black hole; (b) suppressed or absent accretion luminosity relative

to ambient gas supply; (c) depletion of the loosely bound stellar population over Gyr timescales, observable today as a low-mass stellar mass-function slope in the central ~ 0.1 pc falling below the envelope of collisional N -body models without engineered feeding (González Prieto et al., 2025); and (d) possibly, burst-mode high-energy neutrino emission if micro-black-hole computation of the Dvali and Osmanov (2023) type is employed. These residues define the falsification program of Section 6.

The hypothesis (MTH). *If P1–P3 hold, and migration is achievable at a cost the gradient repays (Section 4.5), then the oldest computation-maximizing civilizations in the Galaxy have already relocated to environments of the P2 class; their absence from planetary systems, the radio spectrum, and waste-heat surveys is a selection effect of the thermodynamic gradient; and the appropriate search strategy is targeted scrutiny of the small set of P2-optimal Galactic systems for the P4 residues.*

Three corollaries sharpen the contrast with neighbouring hypotheses. First, against aestivation: the MTH predicts present activity in P2 environments, not dormancy, so it is not vulnerable to the Bennett et al. (2019) objection. Second, against the dark-forest family of concealment hypotheses: silence here is a by-product of efficiency, not a strategic choice, so the MTH requires no game-theoretic coordination among civilizations. Third, against Smart’s original transcension: every MTH process occurs in ordinary spacetime around an ordinary Kerr black hole, so the hypothesis remains permanently coupled to observation.

3.1 Selection criteria over host systems

P2 implies a ranking of real systems. The optimum is set by competing requirements:

1. **Black-hole mass in the intermediate range** (10^3 – $10^5 M_\odot$). Energy extraction (Section 4.1) scales with available fuel and with Eddington-limited throughput ($\propto M$), favouring larger M . The lower bound is set by tides. The differential field at the Schwarzschild ISCO is $2GM/r_{\text{ISCO}}^3 = c^6/(108 G^2 M^2)$, which is $\simeq 0.95 \text{ s}^{-2}$ at $2 \times 10^4 M_\odot$ and scales as M^{-2} ; requiring a 10-m free-flying node to stay below $\sim 10^2 g$ of differential acceleration gives $M \gtrsim 2 \times 10^3 M_\odot$, of which the stated 10^3 floor is the round order of magnitude. The upper bound is not a cost that grows with M but two considerations that cap the useful range: Bekenstein–Hawking storage already saturates any plausible need at $10^4 M_\odot$ (Section 4.3), so further mass buys nothing on that axis, and the environments in which holes above $10^5 M_\odot$ are actually found are deep galactic-centre potentials crowded with disruptive astrophysics (Section 5.5). The intermediate range secures every advantage at minimal exposure.
2. **High spin, or spin-up feasibility.** The Blandford–Znajek channel and the deep-ISCO efficiencies require $a_\star \gtrsim 0.9$; a hole of modest spin can be spun up by prograde disk accretion at a calculable mass cost (Thorne, 1974), provided fuel exists (criterion 3).
3. **A dense, old, low-mass stellar reservoir.** Sustained Eddington-limited feeding of a $\sim 2 \times 10^4 M_\odot$ hole consumes of order one solar mass per two millennia (Section 4.1); at ω Cen’s central density the inner parsec supplies 3.7×10^4 stars and the central ~ 10 pc of order 4.3×10^6 (Table 5), at relative velocities of tens of km s^{-1} . The contested status of this fuel is era-relative: it is uncontested by fusion-era economics, but under star lifting the same M dwarfs are prize fuel (fully convective, hence hydrogen-complete, with $\sim 10^{12}$ -yr lifetimes; Scoggins and Kipping 2023), so among computation-maximizing lineages the reservoir is maximally contested, an early-arrival premium developed in the companion economics paper.

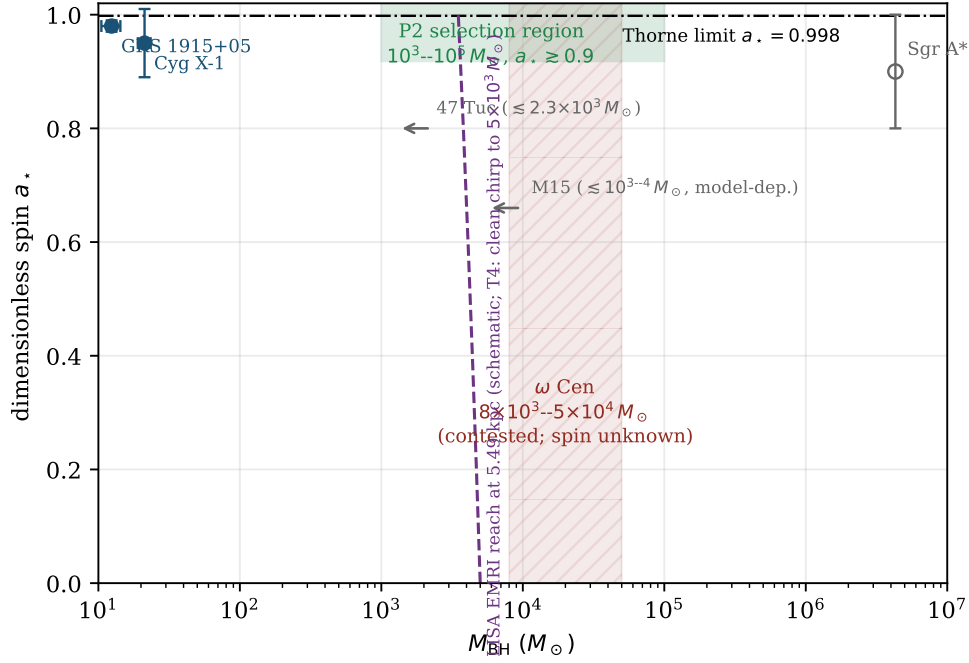


Figure 2: The mass–spin plane. Blue: stellar-mass black holes with spin measurements (representative continuum-fitting values; compilation in [Reynolds 2021](#)). Red hatched band: the contested ω Cen central mass, 8×10^3 – $5 \times 10^4 M_\odot$ (Table 6), spin unknown. Grey arrows: IMBH candidates constrained only from above (47 Tuc, [Kızıltan et al. 2017](#); M15, [Greene et al. 2020](#)). Grey open symbol: Sgr A* (spin inferred; [GRAVITY Collaboration 2022](#)). Green: the P2 selection region, 10^3 – $10^5 M_\odot$ at $a_\star \gtrsim 0.9$ (criteria 1–2). Dash-dotted line: the [Thorne \(1974\)](#) spin-equilibrium limit $a_\star = 0.998$. Dashed curve: schematic LISA EMRI mass reach at 5.49 kpc, anchored at the T4 threshold of $8.2 \times 10^3 M_\odot$ (Table 7) and drawn slightly deeper at high spin, where the inspiral spends more cycles in band. The curve states a requirement the test places on LISA, not a demonstrated reach at this distance; no mission-level sensitivity study for cluster EMRIs at 5.49 kpc is cited here, and the threshold should be read as the mass resolution T4 would need.

4. **Dynamical age and quiescence.** A system $\gtrsim 10$ Gyr old has been reachable by any Galactic civilization arising in the first stellar generations; a relaxed, gas-poor environment minimizes both natural accretion noise and hazards to infrastructure.

Milky Way globular clusters with massive-IMBH candidates satisfy all four criteria simultaneously; ω Cen satisfies them maximally (Section 5). The Galactic Centre fails criterion 1 and 4; isolated stellar-mass black holes fail criterion 3; young massive clusters fail criterion 4. Retention is not the bottleneck: survival analyses of IMBHs in dense clusters find that holes above $\sim 10^3 M_\odot$ are generally retained against dynamical ejection over a Hubble time ([Martinez et al., 2026](#)), so if formation occurs at all, the P2 class is populated today. Figure 2 locates the measured-spin stellar-mass holes, the IMBH candidates, and the P2 selection region in the mass–spin plane.

Because a selection theory that never exhibits its selection invites the charge of unfalsifiability by retreat, Table 1 scores the leading Galactic candidates against the four criteria. The ranking is coarse (the IMBH-evidence column in particular is contested for every entry), but it is the operational content of P2: the class-level kill statement of Section 7 quantifies over exactly these systems.

Table 1: The P2 selection criteria applied to the leading Galactic candidate systems. Symbol columns (current IMBH evidence; inferred or claimed mass versus the preferred 10^3 – $10^5 M_\odot$ range; low-mass fuel reservoir; dynamical age and quiescence): ++ strong, + favourable, ○ neutral or contested, – unfavourable. Quantitative columns: distance and central mass density from [Harris \(1996\)](#); [Baumgardt and Vasiliev \(2021\)](#), rounded to leading digit; $r_{\text{infl}} = GM/\sigma_c^2$ evaluated at a reference $10^4 M_\odot$ and the catalogued central dispersion, which for ω Cen is $\sigma_c = 16.8 \text{ km s}^{-1}$ ([Harris, 1996](#)) and gives 0.15 pc; earlier drafts printed 0.10 pc for this row, which corresponds to $\sigma_c = 20.7 \text{ km s}^{-1}$ and did not match the caption’s own convention. Values assume an isotropic, spherical central-dispersion estimate and carry ~ 50 per cent systematics from that assumption alone; best current central-mass constraint with source.

System	Ev.	Mass	Fuel	Age	d (kpc)	ρ_c ($M_\odot \text{ pc}^{-3}$)	r_{infl} (pc)	Best current mass constraint ($M_\odot$)
ω Cen	++	++	++	++	5.5	$\sim 3 \times 10^3$	0.15	$\geq 8.2 \times 10^3$ (kinematic) vs. $< 6 \times 10^3$ (timing); contested (Section 5.3) (Häberle et al., 2024b ; Bañares-Hernández et al., 2025)
47 Tuc	○	+	++	++	4.5	$\sim 2 \times 10^5$	0.3	$\sim 2.3 \times 10^3$ timing claim (Kızıltan et al., 2017); not required by later modelling (Mann et al., 2019)
M54	+	+	+	+	26.3	$\sim 10^5$	0.2	$\sim 10^4$ kinematic hint (Greene et al., 2020); Sgr dwarf nucleus
NGC 6388	○	+	+	+	9.9	$\sim 5 \times 10^5$	0.12	kinematic claims disputed; deep radio null, $\lesssim \text{few} \times 10^3$ (Tremou et al., 2018)
NGC 6441	○	+	+	+	11.6	$\sim 4 \times 10^5$	0.13	as NGC 6388; radio null (Tremou et al., 2018)
Terzan 5	–	○	++	○	6.9	$\sim 3 \times 10^5$	0.2	no direct evidence (heavy bulge extinction); radio null (Tremou et al., 2018)
M15	○	○	++	++	10.4	$\sim 10^{6-7}$ (cusp)	0.25	$\lesssim 10^{3-4}$, model-dependent; kinematics equally fit by remnants (Greene et al., 2020)

4 The thermodynamic and computational case

This section defends postulate P2 quantitatively. Throughout, M is the black-hole mass, $a_\star \equiv Jc/GM^2 \in [0, 1)$ the dimensionless spin, $r_g \equiv GM/c^2$ the gravitational radius, and we evaluate fiducial numbers at $M = 2 \times 10^4 M_\odot$, the geometric midpoint of the kinematic side of the contested ω Cen range, $\sqrt{8.2 \times 10^3 \times 5 \times 10^4} = 2.0 \times 10^4$ (Section 5); the pulsar-timing side would place the midpoint lower, and we use the kinematic interval throughout so that the fiducial is stated rather than selected. All results in this section are standard physics; no claim about extraterrestrial intelligence is made until Appendix A.

4.1 Energy: accretion and spin extraction versus fusion

Thin-disk efficiency. Matter spiralling through a geometrically thin, radiatively efficient disk is released at the ISCO binding energy. For a circular equatorial orbit at radius r around a Kerr hole, the specific energy at the ISCO gives a radiative efficiency ([Bardeen et al., 1972](#); [Novikov and Thorne, 1973](#))

$$\eta(a_\star) = 1 - E_{\text{ISCO}} = 1 - \sqrt{1 - \frac{2}{3} \frac{r_g}{r_{\text{ISCO}}(a_\star)}}, \quad (1)$$

which rises from $\eta = 1 - \sqrt{8/9} \simeq 0.057$ at $a_\star = 0$ ($r_{\text{ISCO}} = 6r_g$) to 0.321 (pure geodesic value; the photon-capture-corrected equilibrium efficiency is $\simeq 0.30$) at the [Thorne \(1974\)](#) radiative spin-equilibrium limit $a_\star = 0.998$, and formally to $1 - 1/\sqrt{3} \simeq 0.423$ as $a_\star \rightarrow 1$ (Fig. 3). Hydrogen fusion releases 0.71 per cent of the rest mass burned, but a real star offers only its hydrogen fraction $X_{\text{H}} \simeq 0.7$ to that channel, so complete fusion of a star yields $\simeq 5 \times 10^{-3}$ of stellar mass; an unassisted main-sequence star processes

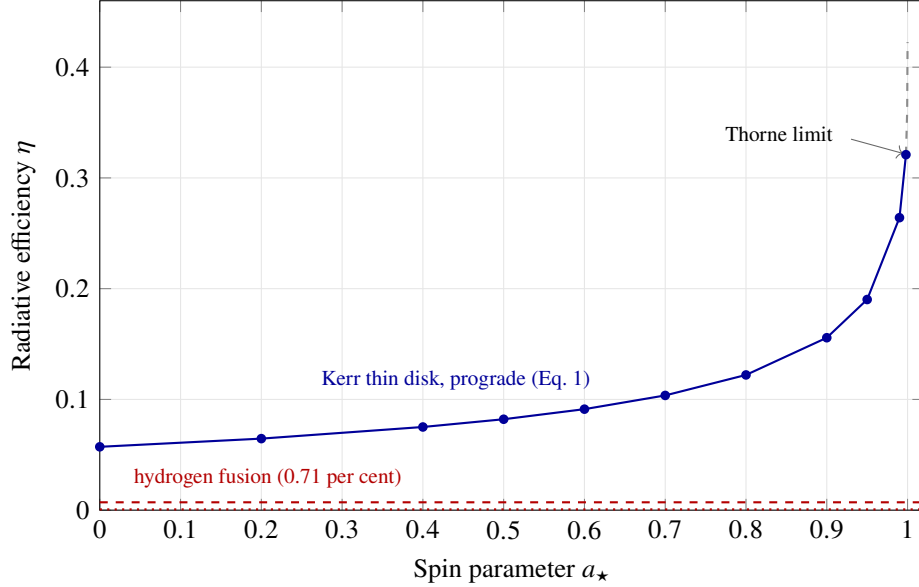


Figure 3: Mass-to-energy conversion efficiency of thin-disk accretion as a function of black-hole spin (Bardeen et al., 1972; Novikov and Thorne, 1973), compared with hydrogen fusion and with the lifetime yield of Dyson-type stellar harvesting. The Thorne (1974) limit $a_* = 0.998$ marks the equilibrium spin reachable by disk accretion alone; the plotted $\eta \approx 0.32$ there is the geodesic value, reduced to ≈ 0.30 once photon capture is included (Thorne, 1974). Curve drawn through exact values of Eq. (1) at the plotted points; the solid branch terminates at the Thorne cap, and the grey dashed continuation to $a_* \rightarrow 1$ marks the formal limit, which radiative spin equilibrium places out of reach.

roughly a tenth of its mass through its core over its lifetime, and only the hydrogen in that tenth burns, so a Dyson-type collector (Dyson, 1960) captures a lifetime yield of order 5×10^{-4} . Both denominators therefore carry X_H , which earlier drafts applied to the stellar yield alone. Per unit of fuel mass under the civilization’s control, disk accretion onto a high-spin hole outperforms complete fusion of the same mass by a factor of ~ 11 at $a_* = 0$ and ~ 60 at the Thorne cap (formally ~ 85 as $a_* \rightarrow 1$), and outperforms Dyson harvesting without star lifting by ~ 115 – 600 (formally ~ 850). Against the hydrogen-only convention used in earlier drafts the Dyson factors read ~ 80 – 420 instead; the numerator is the same, and only the treatment of X_H in the denominator differs. The comparison set matters, because star lifting changes it: a civilization that strips the envelope and feeds the core accesses close to the full hydrogen inventory (Criswell, 1985; Matloff, 2017; Scoggins and Kipping, 2023), raising the yield toward the 5×10^{-3} ceiling and collapsing the Dyson disadvantage to the same ~ 11 – 60 as complete fusion. The entropy-sink and storage advantages (Sections 4.2–4.3) are independent of which factor applies.

Power throughput. The sustainable luminosity is bounded by the Eddington limit,

$$L_{\text{Edd}} = \frac{4\pi G M m_p c}{\sigma_T} \approx 1.26 \times 10^{31} \left(\frac{M}{M_\odot} \right) \text{ W}, \quad (2)$$

giving 1.0×10^{35} , 2.5×10^{35} , and 6.3×10^{35} W at $M = 8.2 \times 10^3$, 2×10^4 , and $5 \times 10^4 M_\odot$ respectively (Frank et al., 2002; Shapiro and Teukolsky, 1983), some 3×10^8 – 2×10^9 times the bolometric output of the Sun across that mass range (6.6×10^8 at $2 \times 10^4 M_\odot$), and hence of any solar Dyson swarm (Fig. 4). The corresponding Eddington accretion rate, $\dot{M}_{\text{Edd}} = L_{\text{Edd}}/\eta c^2$, is remarkably modest: at $\eta = 0.1$ and $M = 2 \times 10^4 M_\odot$, one solar mass sustains the limit for 2,250 years. At the central density of this paper’s

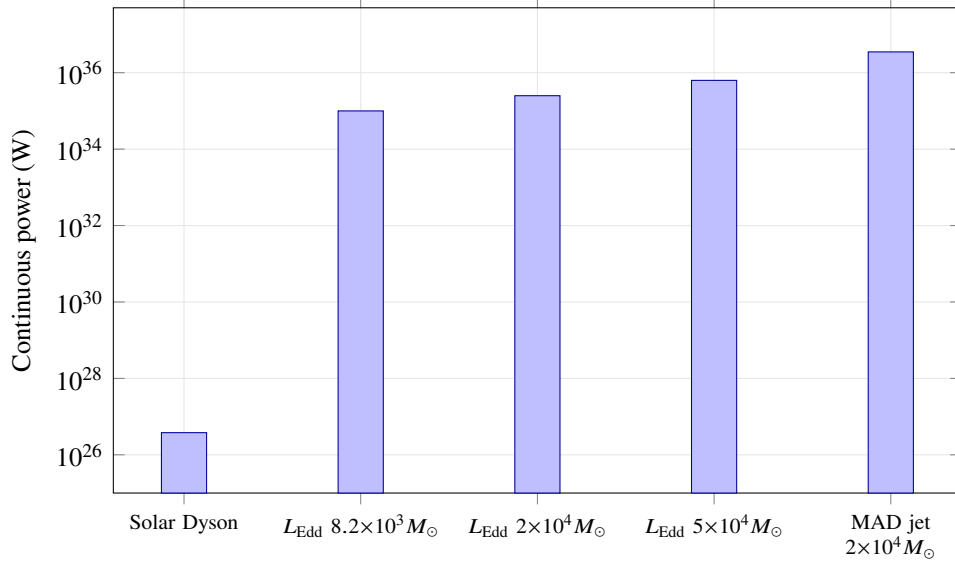


Figure 4: Continuous power available to a civilization from a complete solar Dyson swarm versus Eddington-limited accretion (Eq. 2) onto IMBHs spanning the contested ω Cen mass range, and a magnetically arrested (MAD) Blandford–Znajek jet at $a_{\star} \simeq 0.99$ with $\eta_{\text{jet}} \simeq 1.4$ (Tchekhovskoy et al., 2011), fed at the $\eta = 0.1$ Eddington rate; the two efficiency conventions combine as $P_{\text{jet}} = \eta_{\text{jet}} \dot{M} c^2 = \eta_{\text{jet}} (L_{\text{Edd}}/\eta)$. The IMBH advantage in instantaneous power is roughly nine orders of magnitude.

King model ($\rho_c = 4.0 \times 10^3 M_{\odot} \text{pc}^{-3}$; Section 5.2) the central parsec holds $1.6 \times 10^4 M_{\odot}$, 3.7×10^4 stars, or 3.6×10^7 yr of fuel at fixed hole mass; the working reservoir is the central ~ 10 pc, whose 4.3×10^6 stars amount to 4.2×10^9 yr on the same fixed-mass convention, or 2.0×10^8 yr of elapsed operation once the growing Eddington limit is accounted (Table 5) using bodies (brown dwarfs, low-mass M dwarfs, white dwarfs) that no fusion-based economy values, at the delivery cost priced in Appendix A.

Spin energy and the Blandford–Znajek channel. A Kerr hole stores extractable rotational energy

$$E_{\text{rot}} = \left[1 - \sqrt{\frac{1}{2}(1 + \sqrt{1 - a_{\star}^2})} \right] M c^2, \quad (3)$$

reaching $0.29 M c^2 \simeq 1.0 \times 10^{51}$ J for $M = 2 \times 10^4 M_{\odot}$ at $a_{\star} \rightarrow 1$: a reservoir equal to the entire lifetime fusion output of $\sim 10^7$ Suns, storable indefinitely and tappable on demand. The bracketed quantity is the Christodoulou–Ruffini reducible energy, the difference between M and the irreducible mass M_{irr} (Christodoulou, 1970; Christodoulou and Ruffini, 1971); extraction is reversible only in the limit of vanishing horizon-area increase, and inefficient extraction raises M_{irr} permanently, destroying future capacity, a distinction that matters for the spin-fossil reasoning of Section 6. That this energy is extractable in principle was shown by the ergosphere mechanism of Penrose (1969) and Penrose and Floyd (1971); the astrophysically efficient extraction mechanism is electromagnetic: a horizon threaded by poloidal magnetic flux Φ supplied by an accretion disk acts as a unipolar inductor, driving Poynting-flux jets with power $P_{\text{BZ}} \propto \Phi^2 \Omega_H^2$, where Ω_H is the horizon angular frequency (Blandford and Znajek, 1977). In the membrane paradigm of Thorne et al. (1986) the horizon carries a surface resistivity of 377Ω and the whole configuration reads as a battery with internal resistance, the engineering-facing form of the mechanism and the one relevant to Appendix A’s power-station framing. In the magnetically arrested disk (MAD) regime, general-relativistic magnetohydrodynamic simulations find jet efficiencies $\eta_{\text{jet}} \equiv P_{\text{jet}}/\dot{M} c^2 \simeq 1.4$

at $a_\star \simeq 0.99$ (Tchekhovskoy et al., 2011; Tchekhovskoy, 2015): the jet carries more energy than the accreted rest mass, the excess drawn from spin. The BZ scaling has recently been confirmed from first principles by ab-initio particle-in-cell simulations of Kerr magnetospheres (Meringolo et al., 2025), which additionally identify equatorial magnetic reconnection as a supplementary extraction channel (Comisso and Asenjo, 2021; Camilloni and Rezzolla, 2025); one 2026 preprint reports transient MAD states with jet power exceeding accretion power by over two orders of magnitude (Nathanail, 2026, not yet peer-reviewed). For the MTH it suffices that $\eta_{\text{jet}} \gtrsim 1$ is robust across methods: an engineered MAD configuration around a spun-up IMBH is, per unit fuel, the most efficient large-scale power plant permitted by demonstrated physics.

Spin-up economics. A hole of low natal spin is itself improvable. Prograde thin-disk accretion drives $a_\star \rightarrow 0.998$ after the hole grows by a factor ≈ 2.2 (2.2024) (Bardeen, 1970; Thorne, 1974); for an initial $2 \times 10^4 M_\odot$ hole this raises the hole mass by $2.4 \times 10^4 M_\odot$, which is $2.91 \times 10^4 M_\odot$ of delivered rest mass once the ISCO binding energy is charged along the track ($dM_{\text{hole}} = E_{\text{ISCO}} dM_{\text{rest}}$, giving a ratio of 1.209), and consumes 6.8×10^4 cluster stars at $0.43 M_\odot$. The percentages of Appendix A and the tug masses below are keyed to the hole increment, which is the quantity that leaves the reservoir as bound stellar mass minus what the disk radiates away. At Eddington throughput the episode runs $\sim 3.4 \times 10^7$ yr with the rate evaluated at the mean hole mass during the episode. Integrating instead against the rising efficiency and the rising Eddington limit together, $dt = \eta(a_\star) c^2 dM / L_{\text{Edd}}(M)$, gives $\int \eta d \ln M = 0.1179$ and an episode of 5.3×10^7 yr, a factor 1.57 longer, which we carry as the conservative bracket. Earlier drafts printed $\sim 10^8$ yr here; that figure came from a spin-up integrator carrying four compounding errors and is withdrawn.¹ The investment multiplies all subsequent per-unit-fuel returns by ~ 5 (Fig. 3) and unlocks the BZ reservoir. The corollary for observers: spin is a key parameter of the system’s history (Section 6), since the MTH requires engineered systems to approach near-extremal spin; the converse inference is weaker, because some natural formation channels, in particular coherent disk growth in a gas-rich nucleus, also reach high spin (Section 6; Berti and Volonteri, 2008; Reynolds, 2021).

4.2 Entropy disposal: the horizon as the ultimate cold reservoir

Landauer’s principle sets the minimum thermodynamic cost of irreversible computation: erasing one bit dissipates at least $k_B T \ln 2$ into a reservoir at temperature T (Landauer, 1961; Bennett, 1982; Parrondo et al., 2015), a bound verified experimentally (Bérut et al., 2012). Reversible logic evades the bound for intermediate steps (Bennett, 1973; Toffoli, 1980; Fredkin and Toffoli, 1982; Frank, 2002; Athas et al., 1994), but error correction, measurement, and memory reuse impose a residual erasure budget on any real computer (Wolpert, 2019). For a conventional deep-space computer the reservoir floor is the CMB at $T_\gamma = 2.7$ K, so the marginal cost is $k_B T_\gamma \ln 2 = 2.6 \times 10^{-23}$ J per bit. This is the floor that aestivation proposes to lower by waiting $\sim 10^{12}$ years (Sandberg et al., 2016).

An event horizon lowers it in the present epoch. By the generalized second law (GSL), the sum of black-hole entropy $S_{\text{BH}} = k_B c^3 A / 4G\hbar$ and exterior entropy is non-decreasing, and entropy carried across the horizon is accounted by the area increase (Bekenstein, 1973, 1974; Bousso, 2002). A computation

¹The shipped integrator (figs/fA_calc7.v1.py) evaluated z_2 as $\sqrt{3z_1^2 + a^2}$ rather than the $\sqrt{3a^2 + z_1^2}$ of Bardeen et al. (1972), took the + root of $r_{\text{ISCO}} = 3 + z_2 \mp \sqrt{(3 - z_1)(3 + z_1 + 2z_2)}$, which is the retrograde orbit, omitted the $1/E_{\text{ISCO}}$ in $da_\star/d \ln M = (\ell - 2a_\star E)/E$, and integrated the duration against dm rather than $d \ln m$. The first two put $\eta(0)$ at 0.0505 and $\eta(0.998)$ at 0.0515, so the figures and tabulated efficiencies in this paper were not produced by that code, which is how the error survived. The repaired routine returns $\eta(0) = 0.05719$, $\eta(0.998) = 0.32099$, $r_{\text{ISCO}}(0.998) = 1.23697 r_g$, reaches $a_\star = 0.998$ at $M_f/M_i = 2.20240$, the growth factor quoted above, and reproduces the $\sqrt{6}$ of Bardeen (1970) as $a_\star \rightarrow 1$. A regression gate now pins all five.

platform orbiting the hole may therefore dump its waste entropy into the horizon rather than radiating it to the sky. The marginal energy cost of horizon disposal is set by the horizon temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} \simeq 6.2 \times 10^{-8} \left(\frac{M_\odot}{M} \right) \text{K}, \quad (4)$$

(Hawking, 1974, 1975): $T_H \simeq 3 \times 10^{-12}$ K at $2 \times 10^4 M_\odot$, twelve orders of magnitude below the CMB. A bit dumped at the horizon costs $k_B T_H \ln 2 \sim 3 \times 10^{-35}$ J against 2.6×10^{-23} J for CMB radiation (Fig. 5). In practical terms the erasure cost ceases to be the binding constraint on computation; the budget shifts to transport of waste heat to the horizon (radiative or conductive tethering from the ISCO swarm inward), an engineering loss rather than a thermodynamic floor. The quantified gain from this substitution is the $\sim 10^{12}$ temperature ratio between the CMB and the horizon; the larger $\sim 10^{30}$ factor cited by Sandberg et al. (2016) assumes cooling to temperatures far below the horizon temperature used here and is not directly comparable. The substitution's decisive property is that it operates without waiting, and therefore without forfeiting the free energy that Bennett et al. (2019) showed dormant civilizations irretrievably lose. The related configuration of Opatrný et al. (2017), a shell absorbing CMB radiation as its hot reservoir and rejecting entropy to a central black hole, demonstrates the same thermodynamic asymmetry in a fully worked classical setting.

The claim should also be stated in its energy-conservation form, because that is where the mainstream objection lands. Curtis et al. (2026) state the standard position: Dyson-scale computation must reradiate nearly all of the energy it absorbs, so mid-infrared thermal emission is the robust technosignature of black-hole-hosted intelligence. For a horizon-sink architecture the premise fails, and it fails on energy grounds rather than entropy grounds: the waste carriers themselves cross the horizon, and their energy goes with the entropy. Writing f_{sink} for the fraction of waste *energy* delivered into the capture cone (a vocabulary adopted throughout this series), only the missed fraction $(1 - f_{\text{sink}})$ must be reradiated, at swarm temperature; the absorbed remainder raises M and thereby lowers T_H further, so the channel is weakly self-improving. The disagreement with Curtis et al. (2026) is thus quantitative and falsifiable: the predicted mid-infrared residue scales with $(1 - f_{\text{sink}})$ rather than with the full computational throughput.

The $k_B T_H \ln 2$ figure is the marginal cost of *disposal at the horizon*; it is not the delivered cost from an orbiting platform, and the gap between the two can be estimated rather than merely flagged. Consider radiative delivery, the least demanding channel. Efficient absorption requires a carrier wavelength $\lambda \lesssim r_g$, the geometric-optics condition for capture rather than diffraction around the hole (aiming is a separate and easier requirement, treated below). At $M = 2 \times 10^4 M_\odot$, $r_g \simeq 3 \times 10^7$ m, so the vacuum-gravitational floor on the carrier energy is $E_\gamma \gtrsim \hbar c / r_g \simeq 7 \times 10^{-33}$ J, corresponding to a frequency of ~ 10 Hz. That floor is unreachable through plasma: propagation requires $\nu > \nu_p = 8.98 \text{ kHz} \sqrt{n_e / \text{cm}^{-3}}$, and even the cluster's ambient $n_e = 0.23 \text{ cm}^{-3}$ (47 Tucanae analogy value; Abbate et al. 2018) gives $\nu_p \simeq 4.3 \text{ kHz}$, while any fed inner-flow environment ($n_e \sim 10^4\text{--}10^8 \text{ cm}^{-3}$ inside $\sim 10^2 r_g$) gives $\nu_p \simeq 1\text{--}90 \text{ MHz}$. The minimum propagating carrier energy $E = \hbar \nu_p$ therefore spans 2.9×10^{-30} J at the ambient density to 6.0×10^{-26} J at the densest fed flow, 2.6 to 6.9 decades above $\hbar c / r_g$. Two consequences follow. First, single-bit-per-photon delivery no longer suffices: at the plasma-cutoff carrier energies the single-bit advantage over the CMB would erode to 9×10^6 in the ambient medium and to 4×10^2 in the densest fed flow considered. What preserves the realized range is dense coding: a carrier may bear many bits, up to the Bekenstein channel capacity (Bekenstein, 1981; Bousso, 2002), and coding at that capacity restores the delivered cost per bit to a factor of $10^6\text{--}10^9$ below the CMB figure of 2.6×10^{-23} J.

This requirement can be stated as a length rather than assumed. The Bekenstein bound caps the information carried by a packet of energy E confined to radius R at $I \leq 2\pi E R / (\hbar c \ln 2)$ (Bekenstein, 1981), so a diffraction-limited packet one wavelength long carries at most $2\pi \hbar / (\hbar c \ln 2) \simeq 57$ bits whatever

Table 2: Delivery loss budget for the radiative channel, and the packet coherence length each budget requires. The losses are the ones named in the text: error-correction overhead, capture and pointing efficiency into the 6.75×10^{-4} solid-angle target, transmission through the swarm without re-interception, and the ≈ 1 per cent lapse correction at $10^2 r_g$. Throughput is their product; required length scales as its inverse. The final column fixes the length at 100 km and reports the realized gain that follows.

Budget	Code rate	Capture f_{sink}	Swarm transm.	Lapse	Through-put	L for $G = 10^6/10^9$	G at $L = 100$ km
Lossless (bound)	1.00	1.00	1.00	1.00	1.000	0.135 / 135 km	7.4×10^8
Optimistic	0.90	0.90	0.90	0.99	0.722	0.19 / 190 km	5.4×10^8
Conservative	0.50	0.30	0.30	0.99	0.045	3.0 / 3.0×10^3 km	3.3×10^7

its frequency. Reaching a realized gain G against the CMB figure requires carriers averaging $b = GE_{\text{carr}}/(k_B T_\gamma \ln 2)$ bits, so the packet coherence length the coding demands is $L = G h c / (57 k_B T_\gamma \ln 2)$, the same at every carrier floor: about 1.4×10^2 m at the bottom of the claimed range and 1.4×10^5 m at its top. Two properties of that expression should be stated plainly, because earlier drafts stated the second one backwards. First, E_{carr} cancels, which is why the length is common to all four floors. Second, these are the lengths at which the packet sits exactly on the Bekenstein bound: at every G and every floor the ratio b/N_λ comes out at 57 bits per wavelength exactly. There is no headroom anywhere in the range, and a code running at a fraction f of capacity needs a packet longer by $1/f$. What the assumption consumes is packet coherence rather than any exotic code, and the quantity to check is therefore the length, not a fraction of capacity.

Delivery losses enter the same way, since a loss of throughput f and a code rate f are the same multiplier on the required length. Table 2 prices the losses the preceding paragraphs name, and reports the G that follows rather than assuming it. Under an optimistic budget the range is reached at 0.19–190 km of coherence; under a conservative one at $3.0\text{--}3.0 \times 10^3$ km. Even the worst corner sits $\sim 10^3$ times inside the $10^2 r_g$ working boundary of 3×10^6 km, so the $10^6\text{--}10^9$ range survives being priced. What does not survive is the earlier claim that it is reached with capacity to spare.

Matter carriers relax the coherence constraint, at the price of an accounting question the earlier draft did not face. A one-gram pellet at 100 km s^{-1} , confined to 1 cm, has a Bekenstein capacity of 2.6×10^{38} bits. Charged with kinetic energy alone it delivers at 1.9×10^{-32} J per bit, a gain of 1.3×10^9 ; but the capacity in the numerator is set by the pellet’s total energy, of which rest mass is 1.8×10^7 times the kinetic part, so charging only the kinetic term is not like-for-like. Charging the rest energy symmetrically gives a gain of 74. Charging instead the opportunity cost of the rest mass, $\eta m c^2$ at $\eta = 0.30$, which is what the civilization forgoes by spending the pellet as a carrier rather than as fuel, gives 247. We quote the matter channel at $\sim 10^2$ and take the 10^9 top of the delivered range from the radiative channel alone.

Second, matter carriers evade the cutoff entirely: cold mass dropped down the capture cone crosses the horizon with its entropy and energy aboard and no dispersion relation to satisfy, and serves as the fallback channel wherever the local plasma forbids cheap photons. Charging the engineering losses of Table 2 against the radiative channel leaves a defensible realized gain of $\sim 10^6\text{--}10^9$ over the 2.7 K sky, at the coherence lengths tabulated there. Gravitational bookkeeping does not change this range: energy accounted at a platform of radius $r \gtrsim 10^2 r_g$ is redshifted at infinity by the platform’s lapse factor, a ≈ 1 per cent correction at these radii. The realized economy therefore sits well above the $k_B T_H \ln 2$ floor and well below the CMB figure, and we quote $10^6\text{--}10^9$ rather than the floor-to-floor ratio of $\sim 10^{12}$ wherever the realized gain is meant. An improvement of six orders of magnitude over the 2.7 K sky already dominates the aestivation trade.

Delivery geometry can be derived rather than assumed. The photon-capture cross-section of a

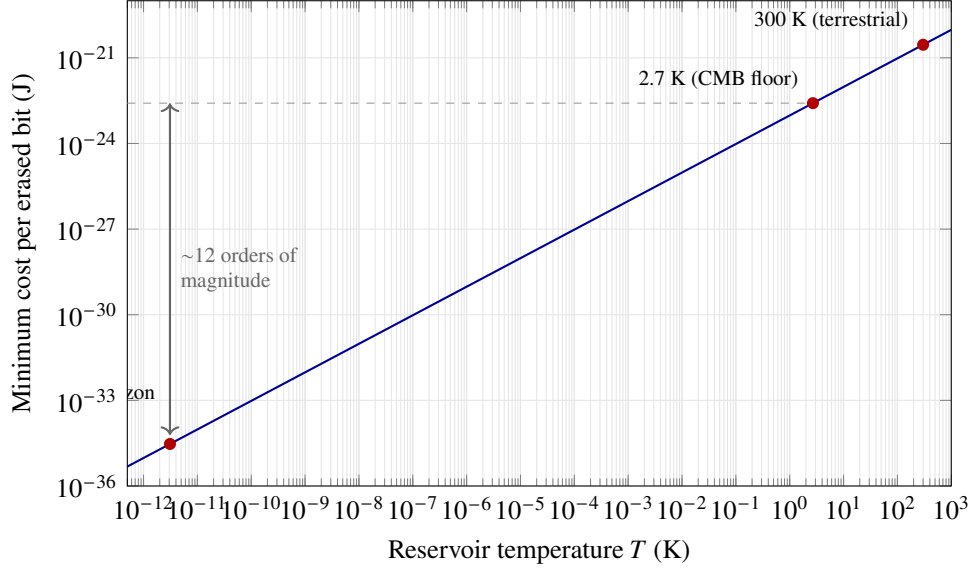


Figure 5: Landauer cost per erased bit, $k_B T \ln 2$, as a function of reservoir temperature. Dumping waste entropy across an IMBH horizon (marginal temperature T_H , Eq. 4) rather than radiating to the CMB lowers the erasure floor by $\sim 10^{12}$, in the present epoch, consistent with the generalized second law (Bekenstein, 1974). The aestivation hypothesis (Sandberg et al., 2016) sought a comparable-in-kind gain by waiting $\sim 10^{12}$ years for cosmological cooling, to still lower assumed reservoir temperatures.

Schwarzschild hole is $\sigma = \pi b_{\text{crit}}^2 = 27\pi r_g^2$, with critical impact parameter $b_{\text{crit}} = 3\sqrt{3} r_g$; from a platform at $r = 10^2 r_g$ the capture target subtends a solid-angle fraction $\Omega_c/4\pi = 6.75 \times 10^{-4}$, so isotropic emission delivers less than a thousandth of its power to the horizon. Reaching $f_{\text{sink}} \rightarrow 1$ therefore requires a beaming gain $\gtrsim 1.5 \times 10^3$, which is optically trivial: the capture cone has half-angle $\theta_c \approx 0.052$ rad, so diffraction demands only an aperture $D \gtrsim 23\lambda$, and the étendue (concentration) limit $1/\sin^2 \theta_c \approx 3.7 \times 10^2$ puts the optic area at a few hundred times the radiator area. (The $27\pi r_g^2$ figure is the Schwarzschild value; the near-extremal prograde equatorial cross-section is somewhat smaller and anisotropic, an $O(1)$ correction.) The binding term in the delivery budget is re-interception within the swarm rather than aiming or aperture: beamed flux must traverse the swarm itself, and the intercepted fraction scales with the swarm covering fraction (Swanson, 2026a).

4.3 Information storage: the Bekenstein–Hawking archive

The Bekenstein bound limits the information content of any bounded system (Bekenstein, 1981; Casini, 2008); black holes saturate it, with

$$\frac{S_{\text{BH}}}{k_B \ln 2} \simeq 1.5 \times 10^{77} \left(\frac{M}{M_\odot} \right)^2 \text{ bits}, \quad (5)$$

i.e. $\sim 6 \times 10^{85}$ bits at $2 \times 10^4 M_\odot$ (Bekenstein, 1973; Hawking, 1975; Lloyd, 2000). That coefficient is the Schwarzschild value; since the architecture ends near-extremal, we quote one self-consistent pair: at $a_\star = 0.998$ the Kerr horizon area per unit M^2 is $0.53 \times$ Schwarzschild, so a $2 \times 10^4 M_\odot$ hole at that spin stores $\approx 3 \times 10^{85}$ bits, and the factor-2.2 mass growth of the spin-up path (Section 4.1) more than recovers the difference through the M^2 scaling. Information crossing the horizon is scrambled but, by unitarity, not destroyed; retrieval timescales via Hawking radiation are, however, of order the evaporation time,

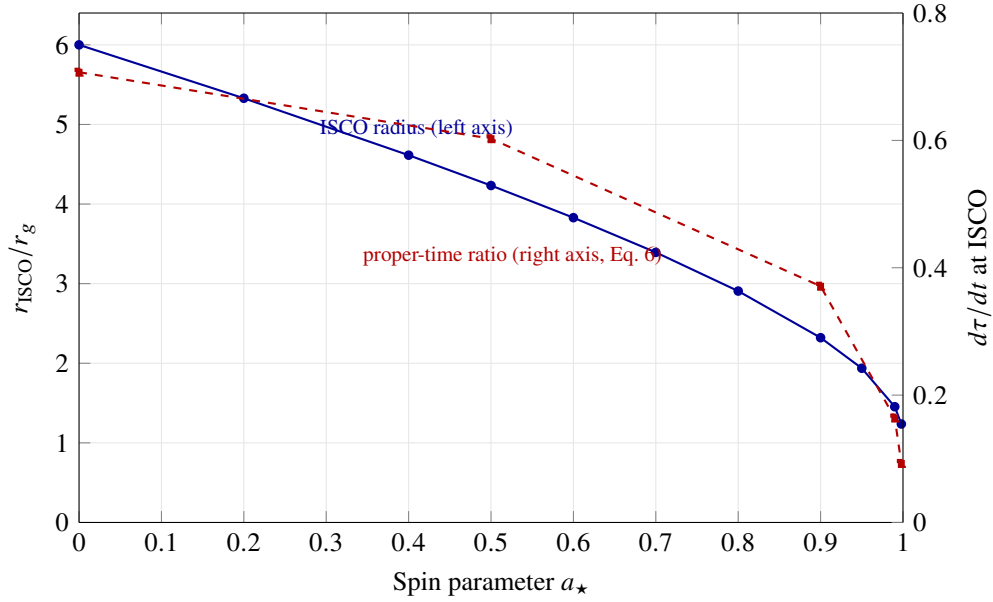


Figure 6: Prograde ISCO radius (left axis) and proper-time ratio for a circular geodesic at the ISCO (right axis) as functions of spin (Bardeen et al., 1972). Spin-up from $a_* = 0$ to the Thorne limit moves the working orbit from $6 r_g$ to $1.24 r_g$ and deepens the available stable time dilation from 1.41:1 to $\sim 11:1$.

$\sim 10^{67} (M/M_\odot)^3$ yr (Page, 1976): the archive is functionally write-only. Black holes likewise saturate the Margolus–Levitin bound on processing rate, $2E/\pi\hbar$ operations per second (Margolus and Levitin, 1998; Lloyd, 2000). We treat these saturation properties as the formal statement that no material technology can out-store or out-compute the object the civilization is already orbiting.

4.4 Time dilation as an engineering parameter

A platform on a circular equatorial geodesic at radius r around a Kerr hole runs slow relative to infinity by

$$\frac{d\tau}{dt} = \frac{\sqrt{1 - 3r_g/r + 2a_*(r_g/r)^{3/2}}}{1 + a_*(r_g/r)^{3/2}}, \quad (6)$$

(Bardeen et al., 1972): $d\tau/dt = \sqrt{1/2} \simeq 0.707$ at the Schwarzschild ISCO, falling to $\simeq 0.093$ (a factor ~ 11) at the prograde ISCO for $a_* = 0.998$ (Fig. 6). Larger factors require powered non-geodesic hovering at diverging thrust cost and are not usable by stable platforms. The $\sim 11:1$ ceiling is set by the Thorne (1974) spin-equilibrium limit $a_* = 0.998$ rather than by orbital stability: the ISCO proper-time ratio falls toward zero as $a_* \rightarrow 1$, and it is the accretion physics capping the spin that caps the dilation. A tiered swarm can therefore place archival and slow-integration processes deep (subjectively skipping across external time) and interactive processes high, with the choice of orbit acting as a clock-rate dial, a resource with no terrestrial analogue, though we note its strategic value (e.g. outwaiting external change) trades directly against responsiveness. The dial is two-way: viewed from outside, a deep tier both runs and receives $11\times$ slower in external time, so deep placement taxes bandwidth to, and coherence with, the shallow tier; and uplink photons arrive blueshifted by the same factor, each depositing $\sim 11\times$ its emitted energy at the deep receiver.

Table 3: One IMBH versus a same-order swarm of stellar-mass holes, priced on the P2 axes. The swarm column takes 2.5×10^4 holes of $10 M_\odot$ (total $2.5 \times 10^5 M_\odot$, the upper end of the Bañares-Hernández et al. 2025 extended component); the IMBH column takes the fiducial $2 \times 10^4 M_\odot$. Storage and tidal entries are quoted as ratios (arb. units).

Figure of merit	One $2 \times 10^4 M_\odot$ hole	2.5×10^4 holes of $10 M_\odot$	Advantage
Aggregate L_{Edd}	2.5×10^{35} W	3.2×10^{36} W	swarm, 13×
Storage, $\sum S_i \propto \sum M_i^2$	4×10^8	2.5×10^6	IMBH, 160×
Sink depth, $T_H \propto M^{-1}$	3.1×10^{-12} K	6.2×10^{-9} K	IMBH, 2000×
ISCO tidal field, $\propto M^{-2}$	1	4×10^6	IMBH, 4×10^6
Latency across working system	ms (one orbit)	μ s per hole; pc-scale between holes	IMBH

4.5 Inward versus outward migration

The expansionist default implicitly maximizes resource acquisition; P1 civilizations maximize computation, and the two diverge. Expansion multiplies harvested mass polynomially in time ($\propto t^3$ for spherical growth at fixed speed) while imposing light-lag decoherence on any shared computation: a civilization spread over 10^3 light-years cannot function as one computer on subjective timescales shorter than millennia. Concentration, by contrast, multiplies the yield per unit mass by the factors of Section 4.1 (up to ~ 600 over Dyson harvesting without star lifting at the Thorne cap), lowers the realized erasure cost by 10^6 – 10^9 (Section 4.2), and keeps the entire system within light-seconds of itself. We must be careful about what this establishes. An undiscounted maximizer of total integrated computation over unbounded time is served by expansion despite the per-unit-mass penalty, because controlled mass grows as t^3 and per-unit efficiency gains are bounded. The gradient points inward only under at least one of the three conditions already carried in P1: (i) temporal discounting, so that near-term computation outweighs the polynomial tail; (ii) a requirement that the computation be unified and latency-bound, so that light-lag decoherence caps the useful radius; or (iii) bounded planning horizons, within which the concentration factors dominate the t^3 growth. Under any of these, beamed-sail transport makes a cluster-hosted IMBH reachable at negligible marginal cost (Lubin, 2016; Armstrong and Sandberg, 2013) and concentration wins. We stress the modesty of the claim actually needed: P1 requires only that some civilizations satisfy one of (i)–(iii) and follow the gradient. Expansionist and stay-at-home civilizations may coexist with migrators; the Fermi-relevant point is that the oldest and most capable optimizers, the very civilizations whose absence the paradox finds most puzzling (Hanson et al., 2021), are the ones the gradient captures.

4.6 One hole rather than many

The comparison set for P2 must include a rival platform that the falsification program itself puts on the table: the dark remnant swarm of hypothesis H_1 (Section 6). Under H_1 the centre of ω Cen contains an extended $2\text{--}3 \times 10^5 M_\odot$ component of stellar-mass black holes (Bañares-Hernández et al., 2025), twelve times the mass of the fiducial $2 \times 10^4 M_\odot$ IMBH. Because $L_{\text{Edd}} \propto M$, the swarm’s aggregate Eddington power is $\sim 3.2 \times 10^{36}$ W against the IMBH’s 2.5×10^{35} W: on raw power the “falsifying” configuration is the better platform by a factor of ~ 13 , and a selection theory that ignored this would be assuming its conclusion. Table 3 prices the two configurations against the P2 axes.

The IMBH wins decisively on three of the four P2 axes and loses on the fourth. Storage scales as $\sum M_i^2$, so fragmenting the central mass into 2.5×10^4 holes of $10 M_\odot$ costs a factor of 160 even against the twelvefold mass advantage. Sink depth scales as $T_H \propto M^{-1}$, a 2000× edge for the IMBH, and the edge

survives the delivery-level accounting of Section 4.2: the realized erasure economy tracks the delivered carrier cost rather than T_H directly, and the vacuum carrier floor hc/r_g is 1.3×10^{-29} J for a $10 M_\odot$ hole against 6.7×10^{-33} J for the IMBH, the same $\sim 2 \times 10^3$ ratio. The ISCO tidal field, $\propto M^{-2}$ at fixed r/r_g , is 4×10^6 times harsher at $10 M_\odot$, excluding all but dust-scale hardware from the deep orbits where the dilation and efficiency gains live. And a computation distributed over the swarm pays parsec-scale light lag between nodes, against milliseconds across a single working orbit, failing the latency-bound condition of Section 4.5 by construction. A computation-maximizer offered both configurations takes the single hole; consolidation is worth a large premium under the same objective.

The premium is a statement about preference, not about reach, and the gap between the two bounds what T3 can carry. Consolidating a remnant swarm means merging stellar-mass holes, and merger products receive gravitational-wave recoil kicks of 10^2 – 10^3 km s $^{-1}$ for generic spin orientations (Campanelli et al., 2007; Lousto and Zlochower, 2011), against a central escape speed of some tens of km s $^{-1}$ in ω Cen. Consolidation by merger therefore ejects most of its own products, and we know of no mechanism by which a civilization could assemble 2.5×10^4 holes into one while retaining them. The subsection establishes that a maximizer would prefer one hole. It does not establish that one could be built.

This derivation does real work in the falsification program. T3 (Table 7) treats a LISA-resolved remnant swarm as a decisive falsifier, and this subsection is why: the swarm loses on storage, sink depth, and tides by two to six orders of magnitude, so a maximizer with gigayears of run of the cluster would not have left the central mass in that configuration. T3 is thereby a derived kill switch rather than a stipulated one, conditional on the preference argument above transferring to a reachability claim, which the recoil problem leaves open. A resolved swarm is decisive against H_2 only if a maximizer with gigayears of run of the cluster could have consolidated it; if consolidation is dynamically blocked, a swarm is evidence about the cluster’s formation history, not about whether anyone was there.

5 Omega Centauri as the optimal observational target

5.1 The cluster

ω Centauri is the most massive Galactic globular cluster ($M_{\text{cl}} \approx 4 \times 10^6 M_\odot$, $\sim 10^7$ stars; Harris 1996), at a kinematic distance of 5.49 ± 0.06 kpc (Häberle et al., 2025), in $\sim 2\sigma$ tension with the Gaia EDR3 parallax (5.24 ± 0.11 kpc; Soltis et al. 2021), whose error budget is dominated by the parallax zero-point systematics discussed by those authors, and consistent with the combined catalogue value (5.43 ± 0.05 kpc; Baumgardt and Vasiliev 2021); nothing in this paper depends on the difference. It is anomalous among globular clusters in nearly every respect that matters here: it hosts multiple stellar populations spanning a broad metallicity range (Johnson and Pilachowski, 2010), a mean stellar age of 12.08 ± 0.01 Gyr with an intrinsic spread of 0.75 Gyr (Clontz et al., 2024), significant rotation, and an associated tidal stream (Ibata et al., 2019); the consensus interpretation is that it is the surviving nucleus of a dwarf galaxy accreted and stripped by the Milky Way (Hilker and Richtler, 2000; Bekki and Tsujimoto, 2019; Souza et al., 2026). Table 4 summarizes the parameters used in this paper.

For the MTH selection criteria of Section 3.1, the stripped-nucleus origin matters twice over. First, nuclear star clusters are the environments most likely to form and retain IMBHs (Greene et al., 2020). Second, the age structure implies that any technological lineage arising anywhere in the progenitor dwarf (or in the early Milky Way) has had ≥ 8 Gyr to locate, reach, and develop the system: at beamed-sail transit speeds of 0.1–0.2 c (Lubin, 2016), crossing the Galaxy takes $\lesssim 10^6$ years, roughly eight thousand times shorter than the available window.

See Paper F App. A for the shared-parameter table across the set, which cross-references this table’s distance, cluster mass, and central-density rows against the other papers’ own choices.

Table 4: Adopted parameters for ω Centauri (NGC 5139).

Quantity	Value	Source
Distance	5.49 ± 0.06 kpc	Häberle et al. (2025)
Cluster mass	$\approx 4 \times 10^6 M_\odot$	Harris (1996); Baumgardt and Vasiliev (2021)
Stellar count	$\sim 10^7$	Harris (1996)
Mean stellar age	12.08 ± 0.01 Gyr (spread 0.75 Gyr)	Clontz et al. (2024)
Origin	stripped dwarf-galaxy nucleus	Hilker and Richtler (2000); Ibata et al. (2019); Souza et al. (2026)
Central density (catalogue)	$\sim 3 \times 10^3 M_\odot \text{pc}^{-3}$	Baumgardt and Vasiliev (2021); cf. $\sim 10^3$ in Pryor and Meylan (1993)
Central density (King model, adopted)	$4.0 \times 10^3 M_\odot \text{pc}^{-3}$	Section 5.2; used for every enclosed-mass figure
Declination	-47.5°	Harris (1996)

Table 5: Stellar enclosed mass from the King profile of Section 5.2. Star counts convert at $0.43 M_\odot$ per star. The two duration columns answer different questions and earlier drafts printed only the first. Column 4 expresses the reservoir as Eddington-years at a hole held fixed at $2 \times 10^4 M_\odot$ (2,250 yr per solar mass, Section 4.1); it is linear in fuel mass and states the reservoir size; it is not an elapsed time. Column 5 is the elapsed time to actually consume the reservoir, $t_S \ln(1 + M_{\text{fuel}}/M_i)$ with $t_S = \eta c^2 / (L_{\text{Edd}}/M) = 4.5 \times 10^7$ yr at $\eta = 0.1$, which accounts for the Eddington limit rising as the hole grows. The 2-pc and 10-pc rows are bookkeeping rather than scenarios: their endpoint masses, 1.3×10^5 and $1.9 \times 10^6 M_\odot$, leave the 10^3 – $10^5 M_\odot$ range of selection criterion 1, so a hole that consumed either reservoir would no longer be the object this paper selects.

r	$M(< r)$ ($M_\odot$)	stars	Fuel at fixed M (Eddington-yr)	Elapsed (growth-aware)	Endpoint M ($M_\odot$)
1 pc	1.6×10^4	3.7×10^4	3.6×10^7	2.6×10^7 yr	3.6×10^4
2 pc	1.1×10^5	2.5×10^5	2.4×10^8	8.3×10^7 yr	1.3×10^5
10 pc	1.9×10^6	4.3×10^6	4.2×10^9	2.0×10^8 yr	1.9×10^6

5.2 The enclosed-mass profile

The fuel-census figures used in this paper (Sections 3.1 and A) are referred throughout to one stellar enclosed-mass profile, stated here so that every ratio is reconstructible. We integrate a single-mass lowered-isothermal (King 1962) model to the catalogued concentration $c = \log_{10}(r_t/r_c) = 1.31$ and core radius $r_c = 2.37'$ (Harris, 1996), scaled at our adopted distance (Table 4), and normalize to the total cluster mass $4 \times 10^6 M_\odot$. This gives $W_0 = 6.22$, $r_t = 77$ pc, and a central density $\rho_c = 4.03 \times 10^3 M_\odot \text{pc}^{-3}$. That is 34 per cent above the catalogued $\sim 3 \times 10^3$ of Table 4, and the two are not interchangeable: a flat-core estimate of $M(< 1 \text{ pc})$ lands within 6 per cent of the integrated profile at 4.0×10^3 and 20 per cent low at 3×10^3 . Every enclosed-mass, fuel-census and depletion figure in this paper is referred to the model value, and we state it once here rather than quoting the two as though they agreed; the catalogued value is retained in Table 4 as the external anchor the model is scaled against through c and r_c , not through ρ_c . The model's projected half-mass radius, 8.3 pc (5.2'), matches the catalogued 5.00' to 3–4 per cent. Enclosed stellar mass:

The profile is a stellar background and carries no mass segregation; a multi-mass treatment is not needed for any figure quoted in this paper.

5.3 The IMBH candidate and the mass tension

Evidence for a central dark mass in ω Cen has accumulated for two decades, from integrated kinematics (Noyola et al., 2008) through proper-motion modelling that initially yielded only upper limits at the $\sim 10^4 M_\odot$ level (van der Marel and Anderson, 2010), with stellar-mass black-hole subsystems long advanced as the alternative explanation (Zocchi et al., 2019; Breen and Hogg, 2013) and a systematic N -body survey concluding that neither ω Cen nor NGC 6624 required an IMBH (Baumgardt et al., 2019). The situation changed qualitatively when Häberle et al. (2024b), using a proper-motion catalogue of 1.4 million stars built from over 500 HST epochs (Häberle et al., 2024a), identified seven stars within 3 arcsec of the cluster centre moving faster than the local escape velocity, of which five are robust detections. Their velocities on those five stars alone require a point mass $\geq 8,200 M_\odot$; including acceleration limits on the same stars raises the lower bound to $21,100 M_\odot$ at 99 per cent confidence (main text of Häberle et al. 2024b).² Independent N -body modelling finds that an IMBH grown in situ to $\sim 5 \times 10^4 M_\odot$ reproduces the present-day cluster structure and fast-star population (González Prieto et al., 2025).

Against this stands Bañares-Hernández et al. (2025): a joint analysis of stellar kinematics and timing accelerations of the cluster’s millisecond pulsars that places a 3σ upper limit of $\sim 6 \times 10^3 M_\odot$ on any central point mass, favouring instead an extended central dark component of $\approx 2\text{--}3 \times 10^5 M_\odot$, naturally interpreted as a centrally concentrated cluster of stellar remnants. The most recent joint MeerKAT–Parkes timing analysis, by contrast, finds its measurements insensitive to an IMBH in the $10^3\text{--}10^4 M_\odot$ range and places a 90 per cent upper limit of $< 10^5 M_\odot$ on the central mass (Colomo Bernadich et al., 2026), a bound compatible with both camps. The two leading results are formally inconsistent under each side’s stated modelling assumptions, and the disagreement is unresolved at the time of writing (Table 6). We take no side; for this paper, the tension is what makes the system valuable as a target. The MTH requires a genuine IMBH (P2), so the Bañares-Hernández et al. (2025) scenario, if confirmed, falsifies the ω Cen application outright (one of several clean kill switches catalogued in Section 6). LISA can settle the question definitively, contingent on a suitable source being in-band during the mission: a single IMBH undergoing a compact-object inspiral yields resolvable extreme- or intermediate-mass-ratio-inspiral (EMRI/IMRI) signals with percent-level mass and spin measurement, whereas a remnant swarm yields a qualitatively different gravitational-wave signature (Amaro-Seoane, 2018; Babak et al., 2017; Colpi et al., 2024).

The contingency deserves emphasis, because the per-cluster IMRI rate for a $\sim 10^4 M_\odot$ hole is uncertain by orders of magnitude, and the plausible outcome of a 4–10 yr mission is that no inspiral is caught in-band, in which case the gravitational-wave branch of Section 6 returns “inconclusive” rather than a verdict (Fig. 7, silence branch). Two mitigations exist. The dense fast-star environment resolved by Häberle et al. (2024b) raises the prior on close compact companions relative to a field IMBH, though it does not remove the rate uncertainty. More important, an inspiral is not the only LISA channel: Stokov et al. (2022) show that a Galactic-cluster IMBH can reveal itself through Doppler modulation of the gravitational-wave signal of an orbiting stellar-mass binary, a continuous signature that requires no merger event during the mission. This channel constrains the mass, though not the spin, so the decisive T5 measurement remains inspiral-contingent.

One element of the predicted dark population has now been observed directly. Whitaker et al. (2026) report the astrometric detection of a $4.46^{+1.22}_{-1.01} M_\odot$ black hole with reported orbital elements $P = 94$ yr, $e = 0.72$, $a = 31$ AU and a $\sim 0.8 M_\odot$ main-sequence companion,³ from HST astrometry spanning

²The abstract of Häberle et al. (2024b) quotes only the velocity-only $8,200 M_\odot$ limit; the $21,100 M_\odot$ figure appears in the main text. A Nature Author Correction (doi:10.1038/s41586-024-08017-4) fixes an equation typo and a right-ascension offset; the results are unaffected.

³We note without resolution that the reported elements are in mild internal tension: Kepler’s third law with $a = 31$ AU and

Table 6: Current constraints on the central dark mass of ω Cen. The kinematic lower bounds and the pulsar-timing upper bound are mutually inconsistent under stated assumptions; resolution is expected from LISA EMRI observations and continued astrometry.

Constraint	Value	Method / source
Lower bound (velocities only)	$\geq 8,200 M_{\odot}$	5 of 7 fast stars (robust subset), HST proper motions (Häberle et al., 2024b)
Lower bound (with accelerations)	$\geq 21,100 M_{\odot}$ (99%)	same stars, acceleration limits (Häberle et al., 2024b)
N -body growth models	$\sim 4.7\text{--}5.1 \times 10^4 M_{\odot}$	cluster-evolution simulations (González Prieto et al., 2025)
Upper bound (point mass)	$< 6 \times 10^3 M_{\odot}$ (3σ)	stellar kinematics + MSP timing (Bañares-Hernández et al., 2025)
Favoured alternative	extended $2\text{--}3 \times 10^5 M_{\odot}$	same analysis (Bañares-Hernández et al., 2025)
Timing (joint MeerKAT–Parkes)	insensitive at $10^3\text{--}10^4 M_{\odot}$; $< 10^5 M_{\odot}$ (90%)	MSP timing (Colomo & Bernadich et al., 2026)
Historical upper limits	$\lesssim 10^4 M_{\odot}$ (model-dep.)	HST proper motions (van der Marel and Anderson, 2010); see also Zocchi et al. (2019)

2002–2023 combined with JWST imaging: the first confirmed stellar-mass black hole in ω Cen, and the first astrometric detection of one in any globular cluster. The detection confirms that the cluster forms and retains stellar-mass black holes, the population of which both sides of the mass tension require ($\sim 10^4$ such remnants in the extended-component scenario; a formation reservoir in the IMBH-growth scenario), and it anchors the low-mass end of the central mass budget with an object whose parameters are measured rather than modelled.

5.4 Electromagnetic silence

The centre of ω Cen is extraordinarily dark, whatever occupies it. The deepest radio observation of any globular cluster (~ 170 h with ATCA at 7.25 GHz, reaching an rms of $1.1 \mu\text{Jy beam}^{-1}$) detects nothing at any proposed cluster centre, bounding a putative IMBH accreting at the Bondi rate to a model-specific accretion fraction $\lesssim 4 \times 10^{-3}$ (3σ ; model-dependent, with ≈ 1 dex of systematic from the fundamental-plane conversion) under Mahida et al.’s own radiative/jet assumptions (Mahida et al., 2026), not the accretion efficiency $\epsilon \equiv L_{\text{bol}}/(\dot{M}_{\text{Bondi}} c^2)$ (see Paper F’s convention table, Swanson 2026b), the deepest point yet in the radio-limit framework established for globular clusters by the MAVERIC survey (Tremou et al., 2018). JWST NIRCам and MIRI photometry of the central field likewise finds no source with the spectral energy distribution of an accreting IMBH, with limits most constraining at the low-mass end of the allowed range (Chen et al., 2026). The X-ray channel, the standard probe of quiescent accretion, agrees: a ~ 291 ks Chandra exposure detects no central point source, limiting the unabsorbed luminosity to $L_X(0.5\text{--}7.0 \text{ keV}) \lesssim 1.6 \times 10^{30} \text{ erg s}^{-1}$, an Eddington ratio of 1.3×10^{-12} for a $10^4 M_{\odot}$ hole, or 6×10^{-13} at the $2 \times 10^4 M_{\odot}$ fiducial (Haggard et al., 2013). To our knowledge, no dedicated technosignature search of ω Cen has been conducted at any wavelength, although the commensal opportunity is unusually rich:

$M_{\text{tot}} \simeq 5.3 M_{\odot}$ gives $P \simeq 75$ yr; the reported period exceeds this Kepler-implied value by ~ 25 per cent; reconciliation lies within the quoted uncertainties, since the $^{+1.22}_{-1.01} M_{\odot}$ range on the black-hole mass alone moves M_{tot} between 4.3 and $6.5 M_{\odot}$ and hence the Kepler-implied period between 83 and 68 yr at fixed a , while a semi-major axis larger by the same relative amount as the period discrepancy closes the rest.

ω Cen is a flagship MeerKAT pulsar-timing target, so every timing session is a potential commensal SETI epoch on the cluster for Breakthrough Listen’s MeerKAT backend; the first dedicated globular-cluster SETI survey (Huang et al., 2026b, five northern clusters with FAST; ω Cen is inaccessible from FAST’s latitude) demonstrates both the feasibility and the current emptiness of this niche, and a Gaia-based phase-space anomaly metric over 79 Galactic globular clusters, in which NGC 5139 ranks among the dynamically most complex systems proposed for technosignature-oriented follow-up (Huang et al., 2026a), provides independent validation for treating globular-cluster anomalies as search targets.

The logic matters here: the MTH predicts electromagnetic silence (P3–P4), but silence is equally consistent with, and more parsimoniously explained by, a quiescent, gas-starved compact object in a relaxed, gas-poor old cluster, which we adopt as the null hypothesis H_0 throughout. Silence cannot confirm the MTH; only the discriminating residues of P4 can separate the hypotheses (Section 6).

5.5 The case against Sagittarius A*

A natural objection: if bigger is better, the Galactic Centre hosts a black hole of mass $4.3 \times 10^6 M_\odot$ (GRAVITY Collaboration, 2022), imaged directly as a shadow (Event Horizon Telescope Collaboration, 2022), with abundant fuel. Three of the four selection criteria of Section 3.1 disfavour it; the first is thermodynamic, the second and third strategic (per the register of claim 1, Section 1.1). First, the storage and erasure advantages saturate at IMBH scale: nothing a maximizer needs scales usefully beyond $\sim 10^{4-5} M_\odot$, while hazards do. Second, the Galactic Centre is the most dynamically violent environment in the Galaxy (ongoing star formation, supernovae, magnetar flares, stochastic accretion flares, and a dense population of perturbers), the opposite of a stable substrate for Gyr-scale infrastructure; ω Cen’s relaxed core offers the same physics in a far quieter dynamical environment. Third, the claim structure matters, in the era-relative form of criterion 3: the Galactic Centre’s gas and stars are the future fuel of natural astrophysics and of any fusion-era competitor, whereas a globular cluster’s low-mass reservoir is uncontested by fusion-era economics and contested only among computation-maximizing lineages, the population P1 already conditions on; the premium therefore goes to early arrival rather than to contest avoidance. The MTH thus predicts that the preferred class is IMBHs in dense old clusters rather than supermassive holes, a distinctive and testable preference, since it directs attention at objects mainstream SETI has never examined.

6 Falsification framework and observational program

We define the competing hypotheses explicitly, then the tests.

H_0 (**gas starvation; null**). The central object, whether IMBH or remnant swarm, is electromagnetically silent for the ordinary reason: a relaxed, gas-poor, 12-Gyr-old cluster supplies nothing to accrete. Favoured by parsimony and adopted as default.

H_1 (**dark remnant swarm**). No IMBH exists; the central mass is an extended cluster of stellar remnants (Bañares-Hernández et al., 2025; Zocchi et al., 2019; Breen and Hogg, 2013). Falsifies the ω Cen application of the MTH.

H_2 (**MTH**). An IMBH exists and is, or has been, the substrate of a migrated civilization (Appendix A).

Four load-bearing numbers below are taken from the companion infrastructure analysis (Swanson, 2026a): the T2 waste-heat floor $L_{\text{waste}} \gtrsim 10^{-4} P_{\text{comp}}$, the Phase-3 erosion boundary, the $\sim 4 \times 10^{11}$ yr Blandford–Znajek spindown time behind the T5 natal-spin argument, and the $7.6 \times 10^5 L_\odot$ ambient Bondi ceiling

used in Appendix A. That companion is a preprint and is not yet refereed. We mark the thresholds that depend on it provisional, and a reader who declines the companion should treat T2 and T5 as stated without numerical thresholds.

H_0 and H_2 predict identical electromagnetic appearances today (by design, since P3–P4 entail silence), so electromagnetic non-detections cannot distinguish them. They diverge on dynamical and high-energy observables. Table 7 states six tests with thresholds, instruments, and timelines; Fig. 7 arranges the decisive subset as a decision tree.

The matrix divides into two kinds of test, and the division is not the one an earlier draft implied. T1, T2 and T6 are constraint channels: since H_0 and H_2 predict identical electromagnetic appearances, a deeper null in any of them lowers a ceiling on managed throughput and retires no phase, and there is no null threshold at which they adjudicate between the hypotheses. We state the consequence rather than leave it implicit: no accumulation of electromagnetic non-detections, however deep, can falsify the MTH. T3 and T5 are the decisive tests, T4 a ranking demotion, and all three are contingent on a compact-object inspiral occurring in LISA band during the mission. The MTH cannot survive a confirmed remnant swarm or a non-spinning IMBH, except as vacuous speculation about other systems; but whether it is adjudicated at all in this generation depends on an event rate the paper cannot pin down (Section 5.3). Second, the LISA spin measurement is among the most informative numbers obtainable this generation (if an in-band inspiral delivers it; Section 5.3), though both the natural-spin null and the expected yield must be stated carefully. Two natural channels are on the table for ω Cen, and earlier drafts adopted both at once, which left the high-spin branch with no discriminating power in either direction. We adopt one. The paper’s own account of ω Cen’s origin (Section 5) is a gas-rich, accreted dwarf-galaxy nucleus, so the prior we carry is coherent disk growth, which generically reaches high spin (Berti and Volonteri, 2008; Reynolds, 2021); because Blandford–Znajek spindown at ambient fueling operates on $\sim 4 \times 10^{11}$ yr timescales (Swanson, 2026a), such natal spin survives to the present. Under that prior a measured $a_\star \gtrsim 0.9$ is expected on H_0 as well as on H_2 and is not by itself a residue. The alternative channels sharpen T5 rather than blunt it. Runaway stellar collisions build IMBHs of several $10^3 M_\odot$ (Fujii et al., 2024), though those simulations do not report a remnant spin, so the channel leaves $a_\star$ unconstrained, not low. Growth by randomly oriented accretion episodes drives the spin to an equilibrium near $a_\star \approx 0.1$ – 0.3 with fluctuations of ± 0.2 (King et al., 2008; Berti and Volonteri, 2008), which places substantial prior weight at and below the T5 trigger $a_\star < 0.1$, not in the small tail an earlier draft assumed. What we cannot supply is the relative weight of the coherent and chaotic channels for this object, and the quantity the test turns on is therefore that weight, which we cannot supply. A measured $a_\star \gtrsim 0.9$ counts as supporting evidence only in conjunction: the discriminating observation is the conjunction of high spin, electromagnetic silence far below the ambient Bondi supply, and the regulation signature of T1 (P4a–b together). EMRI parameter estimation at LISA delivers spin to better than percent precision for favourable systems (Babak et al., 2017; Amaro-Seoane, 2018). Third, the program is cheap: T1–T2 are archival or piggyback analyses; T6 requires only a pointed monitoring program on an instrument already built for the southern sky (Adrián-Martínez et al., 2016; The KM3NeT Collaboration, 2025), following the high-energy SETI logic of Lacki and DiKerby (2025) and the pipeline practices of standard point-source searches (Aartsen et al., 2020); the partially built ARCA array has already demonstrated discovery-class capability with the ~ 220 -PeV event KM3-230213A, the most energetic neutrino yet observed (The KM3NeT Collaboration, 2025); and T3–T5 are free by-products of LISA’s planned EMRI science (Colpi et al., 2024).

The T6 burst criterion must clear the atmospheric-neutrino background, and an order-of-magnitude estimate shows it does under the stated cuts. Full-detector ARCA is expected to collect of order 10^3 up-going atmospheric muon-neutrino events per year above 10 TeV over the full up-going sky (effective areas and atmospheric fluxes from the KM3NeT 2.0 letter of intent; Adrián-Martínez et al., 2016). A

Table 7: Falsification matrix for the MTH at ω Cen. “Kills” indicates what a positive result rules out; tests marked $\dagger$ are decisive against the hypothesis as a whole rather than against one phase. LISA timeline assumes launch in the mid-2030s (Colpi et al., 2024); T3–T5 are additionally contingent on a compact-object inspiral being in-band during the mission (Section 5.3), with the Doppler-modulation channel of Stokov et al. (2022) providing a mass-only fallback.

#	Observation	Threshold	Instrument	Timeline	Kills
T1	Accretion luminosity from core (limit-tightening)	Current best mid-infrared bolometric-equivalent limit $L_{\text{acc}} \approx 2.5 \times 10^{26} \text{ W}$ ($\approx 10^{-9} L_{\text{Edd}}$) (Chen et al., 2026), versus the deeper band-limited Chandra X-ray bound $L_X(0.5\text{--}7.0 \text{ keV}) \lesssim 1.6 \times 10^{23} \text{ W}$ ($\lesssim 10^{-12} L_{\text{Edd}}$) (Haggard et al., 2013); near-term target one further decade, $\lesssim 2.5 \times 10^{25} \text{ W}$. Supplementary trigger: sustained ($> 1 \text{ yr}$) brightening to $L_{\text{acc}} > 10^{29} \text{ W}$	JWST MIRI; ATCA/MeerKAT; Chandra	now–2030	Constraint channel, not a kill: each deepening null lowers the ceiling on managed accretion throughput without retiring any phase. A brightening detection kills Phase 5 (managed environment). Bounds the accretion SED itself, where T2 bounds reradiated waste heat (Haggard et al., 2013; Mahida et al., 2026)
T2	Mid-IR waste-heat excess	$> 3\sigma$ over stellar-population model, $10\text{--}25.5 \mu\text{m}$ (MIRI imaging limit); tested against the horizon-sink floor $L_{\text{waste}} \gtrsim 10^{-4} P_{\text{comp}}$ (Swanson, 2026a)	JWST MIRI	now–2030	Constraint channel, not a kill: any Dyson-type capture is excluded, and a deepening null tightens the ceiling on Phase 4–5 throughput (currently $P_{\text{comp}} \lesssim 10^{3-4} L_{\odot}$). Because the horizon-sink floor is $L_{\text{waste}} \gtrsim 10^{-4} P_{\text{comp}}$, no null retires managed operation outright; it drives P_{comp} down without bound
T3 $\dagger$	GW signature inconsistent with single point mass	stochastic background or multiple low-mass inspirals from core	LISA	late 2030s–2040s	H_2 entirely (confirms H_1); removes P2 substrate
T4	EMRI point mass below the kinematic floor (demotion, not kill)	clean chirp, $M < 8.2 \times 10^3 M_{\odot}$ at $> 3\sigma$; the trigger is the Häberle et al. (2024b) kinematic lower bound, not the $10^3 M_{\odot}$ class edge	LISA	late 2030s–2040s	Demotes the selection ranking of Section 3.1: ω Cen would no longer be a P2-optimal destination, though an $8 \times 10^3 M_{\odot}$ hole still delivers $\sim 1.0 \times 10^{35} \text{ W}$ (Eq. 2), so H_2 is weakened rather than excluded
T5 $\dagger$	EMRI spin measurement low	$a_{\star} < 0.1$ at $> 3\sigma$	LISA	late 2030s–2040s	The exploitation model: no BZ reservoir, no spin-up history; MTH survives only in unmodified “pre-arrival” form
T6	High-energy neutrino bursts from core	≥ 3 tracks within 1° in 10^{2-3} s , $E \gtrsim 10 \text{ TeV}$, $> 5\sigma$ post-trials, no astrophysical counterpart	KM3NeT/ARCA (ω Cen is up-going from the Mediterranean; dec. -47°)	2026–2035	Constraint channel, not a kill: a null at full sensitivity closes the Dvali and Osmanov (2023) channel only, which the MTH does not require; a detection is consistent with H_2 without proving it

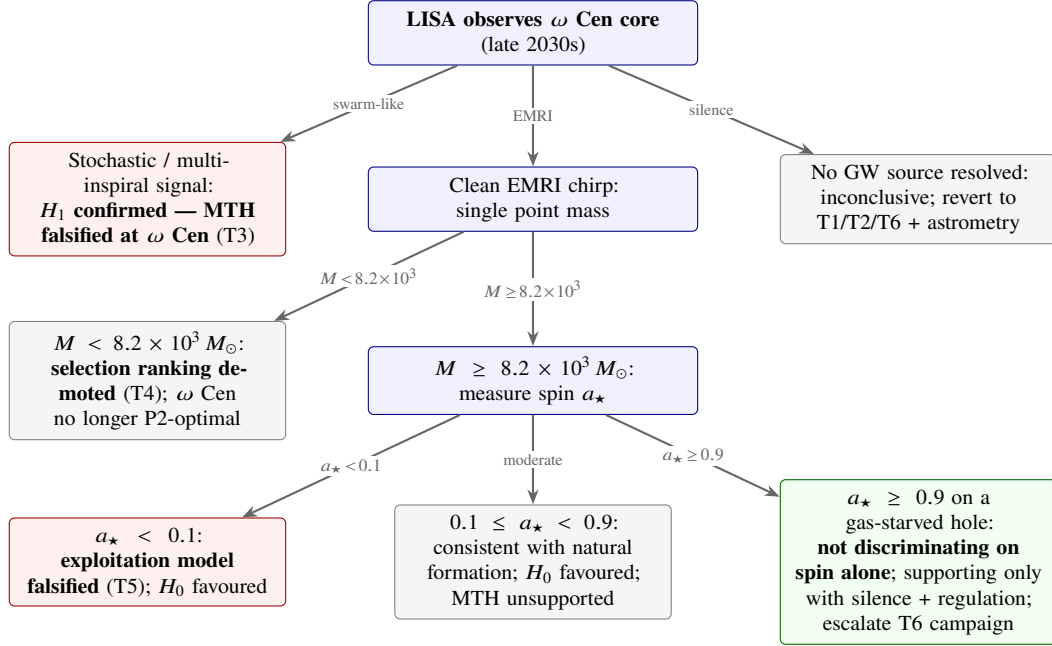


Figure 7: Decision tree for the decisive (gravitational-wave) branch of the falsification program. Red boxes terminate the hypothesis or its exploitation model; the single green branch is the only outcome in which the MTH gains material support, and even there the inference is to “anomaly requiring explanation,” not detection. The mass branches are exhaustive at the $8.2 \times 10^3 M_\odot$ kinematic lower bound of Häberle et al. (2024b), which is the threshold T4 uses; an earlier version split at 5×10^3 on one side and 10^4 on the other, leaving measured masses in $[5 \times 10^3, 10^4]$, the interval containing the kinematic bound itself, with no terminal. Under the coherent-growth prior adopted in the text the high-spin branch is expected on H_0 too, which is why it is labelled as non-discriminating on spin alone.

1° -radius cone subtends $\sim 10^{-3}$ sr, a fraction $\sim 1.5 \times 10^{-4}$ of the up-going hemisphere, giving a source-cone rate of ~ 0.15 events yr^{-1} , i.e. $\lambda \sim 5 \times 10^{-9} \text{ s}^{-1}$. The Poisson expectation in a 10^3 s window is then $\mu \sim 5 \times 10^{-6}$, so the probability of an accidental ≥ 3 -fold multiplet per window is $\mu^3/6 \sim 2 \times 10^{-17}$; a decade of livetime contains $\sim 3 \times 10^5$ independent 10^3 s windows, for an expected false multiplet count of $\sim 10^{-11}$ per decade. The 1° cone is itself deliberately conservative: ARCA’s track angular resolution above 10 TeV is ~ 0.1 – 0.2° , so a resolution-matched cone carries a background lower by a further factor of 25–100. The criterion is background-free by many orders of magnitude even allowing generous uncertainties in the effective area and a trials factor for scanning window durations; the practical limit on T6 is exposure, not background. A dedicated ω Cen technosignature campaign would be the first of its kind for any globular cluster beyond the pilot survey of Huang et al. (2026b), which could not observe ω Cen.

6.1 Supporting evidence criteria

Symmetry requires stating the confirmation side. The MTH gains support only from conjunctions that H_0 finds awkward: (i) a LISA-confirmed IMBH with $a_\star \gtrsim 0.9$ in conjunction with Gyr-scale gas starvation and the regulation signature of T1, since high spin alone admits the natal-disk explanation above; (ii) repeated $> 5\sigma$ neutrino bursts from the core with hard spectra and no electromagnetic counterpart, matching the Dvali and Osmanov (2023) phenomenology; or (iii) a measured low-mass mass-function slope in the central 0.1 pc lying below the González Prieto et al. (2025) model envelope at $> 3\sigma$ (oMEGACat photometry reaches the relevant depth), the concrete form of the P4c depletion residue. Criterion (iii) is the

weakest of the three and we mark it provisional. A 1.3 per cent removal from the 10-pc reservoir preserves the total enclosed mass and is only claimed to be observable because it alters the stellar mass function rather than the density profile; whether a removal of that size imprints a $> 3\sigma$ slope anomaly that survives ~ 12 Gyr of mixing has not been forward-modelled, and until an N -body calculation supplies the predicted amplitude, (iii) states where to look without stating what to expect. No single line, including all three jointly, would constitute proof of technology; they would constitute an anomaly stack justifying escalated scrutiny. Conversely, we commit in advance: if LISA delivers T3, T4, or T5, the ω Cen application of the MTH should be retired without special pleading.

7 Discussion: objections and limits

Parsimony. The MTH posits extraterrestrial intelligence; H_0 posits nothing. By any reasonable prior the null is favoured, and we have structured every comparison accordingly. The defensible claim is that the MTH is cheap to test relative to its information value, not that it is probable: it concentrates the diffuse question “where is everybody?” onto a handful of named objects and a decade-scale instrument schedule, and its decisive observables (IMBH reality, mass, spin) are independently first-rank astrophysics that will be measured anyway.

The anthropic objection. If P1–P2 captured all civilizations, our own existence as a young, loud, planet-bound species is unremarkable; the MTH explains the silence of the old, not the existence of the young. It is therefore not undermined by our own counterexample, but neither can it explain why no expansionist civilizations are visible (cf. [Hanson et al. 2021](#)): if migration is optional, loud lineages should still exist somewhere. The defensible position is that the MTH thins the expected population of loud civilizations by removing its most capable members, sharpening rather than fully dissolving the paradox; full dissolution requires either high migration compliance or supplementary rarity from conventional Drake-equation factors.

Goal stability over 10^8 years. Phase 4 presupposes coherent purpose across timescales that dwarf all institutional experience. We flagged this as the architecture’s largest unargued premise (Appendix A). One partial mitigation: the architecture’s payoff structure is front-loaded (Phase 3 already yields a substrate orders of magnitude beyond planetary computing), so a lineage that fragments mid-program still leaves durable traces, but fewer than we once supposed. The companion analysis ([Swanson, 2026a](#)) finds that an abandoned swarm is not preserved: collisional cascade grinds it to debris within 10^{2-3} yr, and the debris drains into the hole. The durable residues of incoherence are the spin acquired before fragmentation and, if Phase 4 ran long enough, the core depletion it caused; the hardware itself leaves no fossil. The observational program is robust to civilizational incoherence only through those two channels, which is a further reason the dynamical tests (T3–T5) carry the weight.

Tidal and radiative survivability at the ISCO. At $2 \times 10^4 M_\odot$ the tidal field at the Schwarzschild ISCO is $2GM/r_{\text{ISCO}}^3 \simeq 1 \text{ s}^{-2}$: a 10-m node experiences a benign $\sim 1 \text{ g}$ differential, while kilometre-scale rigid structures would face $\sim 10^2 \text{ g}$ — one reason the architecture assumes a swarm of small free-flying nodes rather than a monolithic platform. Radiation from even a managed accretion disk is the harsher constraint, and motivates the tiered-orbit design and sub-Eddington feeding discipline assumed in Appendix A. We have not engineered these systems in detail and do not pretend to; the claim defended is the thermodynamic gradient (P2), not any specific machine.

Why has nothing arrived here? A migration hypothesis must explain why migrators’ probes are not conspicuous in every system, reviving Hart (1975) and Tipler (1980). The MTH answer is economic: for a computation-maximizer, planetary systems are at most transit infrastructure rather than destinations; the Galaxy’s uncollected starlight (an integrated stellar luminosity of $(1-2) \times 10^{37}$ W; cf. Flynn et al. 2006) dwarfs any single system’s output by ten orders of magnitude, and yet (Section 4.5) the marginal value of any outward claim is dominated by deepening the home gravity well. Quiet, minimal, purpose-built transit, rather than occupation, is the predicted footprint, consistent with the null results of artefact SETI to date (Wright et al., 2022). Ivliev (2026) presses the opposite conclusion: once autonomous machine industry matures, outward expansion becomes too cheap and too useful for any civilization to refuse, and the attractor is quiet distributed expansion rather than concentration. We accept the premise that expansion is cheap, and we concede that for an undiscounted maximizer of total integrated computation Ivliev’s inference goes through, since controlled mass grows as t^3 . The disagreement localizes to the auxiliary assumptions of Section 4.5: under temporal discounting, latency-bound unified computation, or bounded planning horizons, expansion does not compete with concentration in yield per unit of controlled resource, and it is civilizations satisfying at least one of those conditions that the MTH captures. The two proposals are nonetheless observationally adjacent, since both predict a Galaxy free of conspicuous macro-engineering, and they diverge where the MTH is testable: Ivliev’s filter predicts no privileged loci, while the MTH concentrates its residues on a short list of named systems.

The kugelblitz weak link. The only positive high-energy signature we predict (T6) inherits the controversy over micro-black-hole formation from radiation (Álvarez-Domínguez et al., 2024; Loeb, 2024). We therefore weight T6 as the cheapest test, not the strongest: its null closes a speculative channel; its detection would be extraordinary but stands or falls with Dvali and Osmanov (2023) physics, not with the MTH core.

Scope of the ω Cen bet. The MTH is a selection theory over a class; ω Cen is its highest-ranked accessible member, not its definition. Falsification at ω Cen (T3, or the ranking demotion of T4) retires the flagship application and substantially weakens the hypothesis (the Galaxy’s best candidate failing is evidence about the class), but formally the theory survives in 47 Tuc, M54, and extragalactic nuclear clusters. We accept the methodological hazard this creates (unfalsifiability by retreat) and therefore bind ourselves to the class-level prediction: *if no Galactic globular-cluster IMBH with $a_\star \gtrsim 0.9$ exists, the MTH is wrong for the Milky Way*. Stating that in deliverable form matters, because a spin measurement requires an inspiral and the per-cluster EMRI rate is uncertain by orders of magnitude (Section 5.3). The settleable version is conditional on yield: if LISA returns spin measurements for ≥ 3 Galactic globular-cluster IMBHs over its mission and none has $a_\star \gtrsim 0.9$, we treat the class prediction as failed. A mission that returns fewer than three leaves the class claim unadjudicated rather than supported, and we commit in advance to reporting it that way, not to reading a null yield as survival.

8 Conclusion

We have argued, first as a matter of established physics, that rapidly spinning massive black holes in dense old stellar clusters constitute the global optimum among present-day environments for long-term computation (by factors of $\sim 10^2$ in energy per unit fuel, $\sim 10^9$ in continuous power, a realized 10^6 – 10^9 in entropy-disposal cost once delivery overheads are charged against the $\sim 10^{12}$ floor-to-floor ratio (Section 4.2), and unbounded factors in archival density), and second, as an explicitly speculative hypothesis, that this gradient acts on the oldest technological civilizations as an attractor whose operation would look, from

outside, like the silence we observe. The Macro Transcension Hypothesis inherits the thermodynamic insight of transcension and aestivation while repairing their respective defects: it needs no exit from the universe and no dormancy, and it survives the [Bennett et al. \(2019\)](#) free-energy objection by importing the cold reservoir into the present rather than waiting for one.

Its principal virtue is observational specificity. The hypothesis selects a named target (Omega Centauri, whose contested 8×10^3 – $5 \times 10^4 M_\odot$ central object, perfect electromagnetic silence, and 12-Gyr head start make it scientifically urgent on entirely conventional grounds) and commits to kill criteria that existing and funded instruments will adjudicate: LISA’s resolution of the IMBH-versus-swarm question and, above all, its spin measurement; deep accretion and waste-heat limits from JWST and southern radio arrays; and a first-of-its-kind neutrino monitoring campaign with KM3NeT. If the central object proves to be a remnant swarm, a light point mass, or a slowly spinning hole, the hypothesis fails at its flagship and we have said so in advance. If instead the 2030s deliver a massive, gas-starved, near-extremally spinning black hole in the quietest old cluster in the sky, the question this paper poses will have earned the right to be taken seriously.

Acknowledgements and disclosure

The author thanks the maintainers of the NASA Astrophysics Data System and arXiv, on which the citation verification for this work relied. **AI assistance disclosure:** drafting, citation verification, derivation checking, and figure preparation for this manuscript were performed with substantial assistance from a large language model (Claude, Anthropic), under the author’s direction; the author reviewed and takes full responsibility for all claims, derivations, and references. Interactive calculators implementing the quantitative material in Sections 4–6 are available as supplementary material at <https://omegacentauri.me>.

Data availability

No new observational data were generated. All cited measurements are available in the referenced publications.

References

- Aartsen, M. G. et al. (2020). Time-integrated neutrino source searches with 10 years of IceCube data. *Physical Review Letters*, 124:051103.
- Abbate, F., Possenti, A., Ridolfi, A., Freire, P. C. C., Camilo, F., Manchester, R. N., and D’Amico, N. (2018). Internal gas models and central black hole in 47 Tucanae using millisecond pulsars. *Monthly Notices of the Royal Astronomical Society*, 481(1):627–637.
- Adrián-Martínez, S. et al. (2016). Letter of intent for KM3NeT 2.0. *Journal of Physics G: Nuclear and Particle Physics*, 43(8):084001.
- Álvarez-Domínguez, Á., Garay, L. J., Martín-Martínez, E., and Polo-Gómez, J. (2024). No black holes from light. *Physical Review Letters*, 133:041401.
- Amaro-Seoane, P. (2018). Relativistic dynamics and extreme mass ratio inspirals. *Living Reviews in Relativity*, 21:4.

- Armstrong, S. and Sandberg, A. (2013). Eternity in six hours: Intergalactic spreading of intelligent life and sharpening the Fermi paradox. *Acta Astronautica*, 89:1–13.
- Athas, W. C., Svensson, L. J., Koller, J. G., Tzartzanis, N., and Chou, E. Y.-C. (1994). Low-power digital systems based on adiabatic-switching principles. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2(4):398–407.
- Babak, S., Gair, J., Sesana, A., Barausse, E., Sopuerta, C. F., Berry, C. P. L., Berti, E., Amaro-Seoane, P., Petiteau, A., and Klein, A. (2017). Science with the space-based interferometer LISA. V. Extreme mass-ratio inspirals. *Physical Review D*, 95(10):103012.
- Bañares-Hernández, A., Calore, F., Martín Camalich, J., and Read, J. I. (2025). New constraints on the central mass contents of Omega Centauri from combined stellar kinematics and pulsar timing. *Astronomy & Astrophysics*, 693:A104.
- Bardeen, J. M. (1970). Kerr metric black holes. *Nature*, 226:64–65.
- Bardeen, J. M., Press, W. H., and Teukolsky, S. A. (1972). Rotating black holes: locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *The Astrophysical Journal*, 178:347–370.
- Barrow, J. D. (1998). *Impossibility: The Limits of Science and the Science of Limits*. Oxford University Press, Oxford.
- Baumgardt, H., He, C., Sweet, S. M., Drinkwater, M., Sollima, A., Hurley, J., Usher, C., Kamann, S., Dalglish, H., Dreizler, S., and Husser, T.-O. (2019). No evidence for intermediate-mass black holes in the globular clusters ω Cen and NGC 6624. *Monthly Notices of the Royal Astronomical Society*, 488:5340–5351.
- Baumgardt, H. and Vasiliev, E. (2021). Accurate distances to Galactic globular clusters through a combination of Gaia EDR3, HST, and literature data. *Monthly Notices of the Royal Astronomical Society*, 505:5957–5977.
- Bekenstein, J. D. (1973). Black holes and entropy. *Physical Review D*, 7:2333–2346.
- Bekenstein, J. D. (1974). Generalized second law of thermodynamics in black-hole physics. *Physical Review D*, 9:3292–3300.
- Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Physical Review D*, 23:287–298.
- Bekki, K. and Tsujimoto, T. (2019). A new formation model for ω Centauri: A complex interplay of astrophysical processes. *The Astrophysical Journal*, 886:121.
- Bennett, C. H. (1973). Logical reversibility of computation. *IBM Journal of Research and Development*, 17:525–532.
- Bennett, C. H. (1982). The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21:905–940.
- Bennett, C. H., Hanson, R., and Riedel, C. J. (2019). Comment on ‘The aestivation hypothesis for resolving Fermi’s paradox’. *Foundations of Physics*, 49:820–829.

- Benson-Tilsen, T. and Soares, N. (2016). Formalizing convergent instrumental goals. In *AAAI Workshop on AI, Ethics, and Society*.
- Berti, E. and Volonteri, M. (2008). Cosmological black hole spin evolution by mergers and accretion. *The Astrophysical Journal*, 684:822–828.
- Bérut, A., Arakelyan, A., Petrosyan, A., Ciliberto, S., Dillenschneider, R., and Lutz, E. (2012). Experimental verification of Landauer’s principle linking information and thermodynamics. *Nature*, 483:187–189.
- Blandford, R. D. and Znajek, R. L. (1977). Electromagnetic extraction of energy from Kerr black holes. *Monthly Notices of the Royal Astronomical Society*, 179:433–456.
- Bostrom, N. (2012). The superintelligent will: Motivation and instrumental rationality in advanced artificial agents. *Minds and Machines*, 22:71–85.
- Bousso, R. (2002). The holographic principle. *Reviews of Modern Physics*, 74:825–874.
- Bradbury, R. J. (1999). Matrioshka brains. Unpublished working manuscript, revised through 2001. Archived at <https://gwnet.net/doc/ai/scaling/hardware/1999-bradbury-matrioshkabrains.pdf>; mirrored at <https://web.archive.org/web/2024/https://gwnet.net/doc/ai/scaling/hardware/1999-bradbury-matrioshkabrains.pdf>.
- Breen, P. G. and Hogg, D. C. (2013). Dynamical evolution of black hole subsystems in idealized star clusters. *Monthly Notices of the Royal Astronomical Society*, 432:2779–2797.
- Breit, G. and Wheeler, J. A. (1934). Collision of two light quanta. *Physical Review*, 46:1087–1091.
- Camilloni, F. and Rezzolla, L. (2025). Self-consistent multidimensional Penrose process driven by magnetic reconnection. *The Astrophysical Journal Letters*, 982:L31.
- Campanelli, M., Lousto, C. O., Zlochower, Y., and Merritt, D. (2007). Maximum gravitational recoil. *Physical Review Letters*, 98:231102.
- Casini, H. (2008). Relative entropy and the Bekenstein bound. *Classical and Quantum Gravity*, 25:205021.
- Chen, S., Hare, J., Kargaltsev, O., Yang, H., Cioffi, D., Häberle, M., and Seth, A. (2026). The intermediate mass black hole in Omega Centauri: Constraints on accretion from JWST. *The Astrophysical Journal*.
- Christodoulou, D. (1970). Reversible and irreversible transformations in black-hole physics. *Physical Review Letters*, 25:1596–1597.
- Christodoulou, D. and Ruffini, R. (1971). Reversible transformations of a charged black hole. *Physical Review D*, 4:3552–3555.
- Ćirković, M. M. (2018). *The Great Silence: The Science and Philosophy of Fermi’s Paradox*. Oxford University Press, Oxford.
- Ćirković, M. M. and Bradbury, R. J. (2006). Galactic gradients, postbiological evolution and the apparent failure of SETI. *New Astronomy*, 11(8):628–639.
- Clontz, C., Seth, A. C., Dotter, A., Häberle, M., Nitschai, M. S., Neumayer, N., Feldmeier-Krause, A., Latour, M., Wang, Z., Souza, S. O., Kacharov, N., Bellini, A., Libralato, M., Pechetti, R., van de Ven, G., and Alfaro-Cuello, M. (2024). oMEGACat. IV. Constraining the ages of Omega Centauri subgiant branch stars with HST and MUSE. *The Astrophysical Journal*, 977:14.

- Colomo i Bernadich, M., Dai, S., Abbate, F., Kerr, M., Bachetti, M., Bhargava, Y., Buchner, S., Johnston, S., Burgay, M., Possenti, A., Nag, R., Ridolfi, A., Carleo, A., Corongiu, A., Freire, P. C. C., Camilo, F., Chen, W., Cadelano, M., Risbud, D., Padmanabh, P. V., Champion, D. J., Kramer, M., Stappers, B., Serylak, M., Balakrishnan, V., Bailes, M., Dutta, A., Vleeschower Calas, L., Venkatraman Krishnan, V., and Men, Y. (2026). A joint MeerKAT and Parkes view of Omega Centauri: New TRAPUM searches and pulsar timing. *arXiv e-prints*, page arXiv:2603.21845.
- Colpi, M. et al. (2024). LISA definition study report. arXiv:2402.07571; ESA-SCI-DIR-RP-002.
- Comisso, L. and Asenjo, F. A. (2021). Magnetic reconnection as a mechanism for energy extraction from rotating black holes. *Physical Review D*, 103:023014.
- Criswell, D. R. (1985). Solar system industrialization: Implications for interstellar migrations. In Finney, B. R. and Jones, E. M., editors, *Interstellar Migration and the Human Experience*, pages 50–87. University of California Press, Berkeley.
- Curtis, O., Adams, V. H., Angerhausen, D., Bates, J., Berea, A., Dick, S. J., Elvis, M., Khatri, S. P., Linares, R., Muco, M., Seager, S., and Wright, J. T. (2026). The Dyson Minds 2025 workshop: SETI around black holes. *Publications of the Astronomical Society of the Pacific*, 138:046001.
- Dvali, G. and Osmanov, Z. N. (2023). Black holes as tools for quantum computing by advanced extraterrestrial civilizations. *International Journal of Astrobiology*, 22(6):617–640.
- Dyson, F. J. (1960). Search for artificial stellar sources of infrared radiation. *Science*, 131(3414):1667–1668.
- Event Horizon Telescope Collaboration (2022). First Sagittarius A* Event Horizon Telescope results. I. The shadow of the supermassive black hole in the center of the Milky Way. *The Astrophysical Journal Letters*, 930:L12.
- Flynn, C., Holmberg, J., Portinari, L., Fuchs, B., and Jahreiß, H. (2006). On the mass-to-light ratio of the local Galactic disc and the optical luminosity of the Galaxy. *Monthly Notices of the Royal Astronomical Society*, 372:1149–1160.
- Forgan, D. H. (2019). *Solving Fermi’s Paradox*. Cambridge Astrobiology. Cambridge University Press, Cambridge.
- Frank, J., King, A., and Raine, D. (2002). *Accretion Power in Astrophysics*. Cambridge University Press, Cambridge, 3rd edition.
- Frank, M. P. (2002). The physical limits of computing. *Computing in Science & Engineering*, 4(3):16–26.
- Fredkin, E. and Toffoli, T. (1982). Conservative logic. *International Journal of Theoretical Physics*, 21:219–253.
- Freitas, Jr., R. A. and Gilbreath, W. P., editors (1982). *Advanced Automation for Space Missions*. Number 2255 in NASA Conference Publication. NASA. Edited volume; proceedings of the 1980 NASA/ASEE Summer Study, University of Santa Clara.
- Frolov, V. P. and Novikov, I. D. (1998). *Black Hole Physics: Basic Concepts and New Developments*. Kluwer Academic, Dordrecht.

- Fujii, M. S., Wang, L., Tanikawa, A., Hirai, Y., and Saitoh, T. R. (2024). Simulations predict intermediate-mass black hole formation in globular clusters. *Science*, 384:1488–1492.
- González Prieto, E., Rodríguez, C. L., and Cabrera, T. (2025). Growing the intermediate-mass black hole in Omega Centauri. *The Astrophysical Journal Letters*, 990:L69.
- GRAVITY Collaboration (2022). Mass distribution in the Galactic Center based on interferometric astrometry of multiple stellar orbits. *Astronomy & Astrophysics*, 657:L12.
- Greene, J. E., Strader, J., and Ho, L. C. (2020). Intermediate-mass black holes. *Annual Review of Astronomy and Astrophysics*, 58:257–312.
- Griffith, R. L., Wright, J. T., Maldonado, J., Povich, M. S., Sigurdsson, S., and Mullan, B. (2015). The $\hat{g}$ infrared search for extraterrestrial civilizations with large energy supplies. III. The reddest extended sources in WISE. *The Astrophysical Journal Supplement Series*, 217(2):25.
- Häberle, M., Neumayer, N., Bellini, A., Libralato, M., Clontz, C., Seth, A. C., et al. (2024a). oMEGACat. II. Photometry and proper motions for 1.4 million stars in Omega Centauri and its rotation in the plane of the sky. *The Astrophysical Journal*, 970:192.
- Häberle, M., Neumayer, N., Clontz, C., Seth, A. C., Bellini, A., et al. (2025). oMEGACat. VI. Analysis of the overall kinematics of Omega Centauri in 3D: Velocity dispersion, kinematic distance, anisotropy, and energy equipartition. *The Astrophysical Journal*, 983:95.
- Häberle, M., Neumayer, N., Seth, A., Bellini, A., Libralato, M., Baumgardt, H., Whitaker, M., Dumont, A., Alfaro-Cuello, M., Anderson, J., Clontz, C., et al. (2024b). Fast-moving stars around an intermediate-mass black hole in ω Centauri. *Nature*, 631(8020):285–288.
- Haggard, D., Cool, A. M., Heinke, C. O., van der Marel, R., Cohn, H. N., Lugger, P. M., and Anderson, J. (2013). A deep Chandra X-ray limit on the putative IMBH in Omega Centauri. *The Astrophysical Journal Letters*, 773:L31.
- Hanson, R., Martin, D., McCarter, C., and Paulson, J. (2021). If loud aliens explain human earliness, quiet aliens are also rare. *The Astrophysical Journal*, 922(2):182.
- Harris, W. E. (1996). A catalog of parameters for globular clusters in the Milky Way. *The Astronomical Journal*, 112:1487.
- Hart, M. H. (1975). An explanation for the absence of extraterrestrials on Earth. *Quarterly Journal of the Royal Astronomical Society*, 16:128–135.
- Hawking, S. W. (1974). Black hole explosions? *Nature*, 248:30–31.
- Hawking, S. W. (1975). Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220.
- Hilker, M. and Richtler, T. (2000). ω Centauri – a former nucleus of a dissolved dwarf galaxy? New evidence from Strömgren photometry. *Astronomy & Astrophysics*, 362:895–909.
- Hsiao, T. Y.-Y., Goto, T., Hashimoto, T., Santos, D. J. D., On, A. Y. L., Kilerci-Eser, E., Wong, Y. H. V., Kim, S. J., Wu, C. K.-W., Ho, S. C.-C., and Lu, T.-Y. (2021). A dyson sphere around a black hole. *Monthly Notices of the Royal Astronomical Society*, 506(2):1723–1732.

- Huang, B.-L., Tao, Z.-Z., and Zhang, T.-J. (2026a). Phase-space crystallization in Galactic globular clusters: A Gaia-based metric and implications for technosignature searches. *The Astrophysical Journal*, 1005:102.
- Huang, B.-L., Tao, Z.-Z., Zhang, T.-J., and Gajjar, V. (2026b). The FAST-SETI Milky Way globular cluster survey. I. A pilot multibeam on-the-fly search of five globular clusters at the L band. *The Astronomical Journal*, 171(1):51.
- Ibata, R. A., Bellazzini, M., Malhan, K., Martin, N., and Bianchini, P. (2019). Identification of the long stellar stream of the prototypical massive globular cluster ω Centauri. *Nature Astronomy*, 3:667–672.
- Inoue, M. and Yokoo, H. (2011). Type III dyson sphere of highly advanced civilisations around a super massive black hole. *Journal of the British Interplanetary Society*, 64:58–62.
- Ivliev, S. (2026). Autonomous AI-cosmoindustry and the quiet expansion filter: A threshold-based resolution of the Fermi paradox. arXiv:2606.13914; preprint.
- Johnson, C. I. and Pilachowski, C. A. (2010). Chemical abundances for 855 giants in the globular cluster Omega Centauri (NGC 5139). *The Astrophysical Journal*, 722:1373–1410.
- Kardashev, N. S. (1964). Transmission of information by extraterrestrial civilizations. *Soviet Astronomy*, 8:217–221.
- King, A. R., Pringle, J. E., and Hofmann, J. A. (2008). The evolution of black hole mass and spin in active galactic nuclei. *Monthly Notices of the Royal Astronomical Society*, 385:1621–1627.
- Kızıltan, B., Baumgardt, H., and Loeb, A. (2017). An intermediate-mass black hole in the centre of the globular cluster 47 Tucanae. *Nature*, 542:203–205.
- Lacki, B. C. and DiKerby, S. (2025). Possibilities for SETI at high energy. arXiv:2506.16351; white paper submitted to the 2025 NASA DARES RFI.
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5:183–191.
- Lingam, M. and Loeb, A. (2021). *Life in the Cosmos: From Biosignatures to Technosignatures*. Harvard University Press, Cambridge, MA.
- Lloyd, S. (2000). Ultimate physical limits to computation. *Nature*, 406:1047–1054.
- Loeb, A. (2024). Comment on “No black holes from light”. arXiv:2408.06714; reply by original authors at arXiv:2408.11097.
- Lousto, C. O. and Zlochower, Y. (2011). Hangup kicks: Still larger recoils by partial spin-orbit alignment of black-hole binaries. *Physical Review Letters*, 107:231102.
- Lubin, P. (2016). A roadmap to interstellar flight. *Journal of the British Interplanetary Society*, 69:40–72.
- Mahida, A. D. et al. (2026). No evidence for accretion around the intermediate-mass black hole in Omega Centauri. *The Astrophysical Journal*, 996:122.

- Mann, C. R., Richer, H., Heyl, J., Anderson, J., Kalirai, J., Caiazzo, I., Möhle, S., Knee, A., and Baumgardt, H. (2019). A multi-mass velocity dispersion model of 47 Tucanae indicates no evidence for an intermediate mass black hole. *The Astrophysical Journal*, 875:1.
- Margolus, N. and Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D: Nonlinear Phenomena*, 120:188–195.
- Martinez, M. A. S., González Prieto, E., and Rasio, F. A. (2026). Survival analysis of intermediate-mass black holes in dense star clusters. *arXiv e-prints*, page arXiv:2602.23431. Submitted to ApJ.
- Matloff, G. L. (2017). Star lifting: An application for alien megastructures. *Journal of the British Interplanetary Society*, 70:458–460.
- Meringolo, C., Camilloni, F., and Rezzolla, L. (2025). Electromagnetic energy extraction from Kerr black holes: Ab initio calculations. *The Astrophysical Journal Letters*, 992:L8.
- Misner, C. W., Thorne, K. S., and Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman, San Francisco.
- Nathanail, A. (2026). Black hole limits redefined: Extreme efficiency in black hole jets. arXiv:2602.22824; preprint, not yet peer-reviewed.
- Novikov, I. D. and Thorne, K. S. (1973). Astrophysics of black holes. In DeWitt, C. and DeWitt, B. S., editors, *Black Holes (Les Houches)*, pages 343–450. Gordon and Breach, New York.
- Noyola, E., Gebhardt, K., and Bergmann, M. (2008). Gemini and Hubble Space Telescope evidence for an intermediate-mass black hole in ω Centauri. *The Astrophysical Journal*, 676:1008–1015.
- Omohundro, S. M. (2008). The basic ai drives. In Wang, P., Goertzel, B., and Franklin, S., editors, *Proceedings of the First AGI Conference (AGI-08)*, pages 483–492. IOS Press.
- Opatrný, T., Richterek, L., and Bakala, P. (2017). Life under a black sun. *American Journal of Physics*, 85(1):14–22.
- Page, D. N. (1976). Particle emission rates from a black hole: Massless particles from an uncharged, nonrotating hole. *Physical Review D*, 13:198–206.
- Parrondo, J. M. R., Horowitz, J. M., and Sagawa, T. (2015). Thermodynamics of information. *Nature Physics*, 11:131–139.
- Penrose, R. (1969). Gravitational collapse: the role of general relativity. *Rivista del Nuovo Cimento*, 1(Numero Speciale):252–276. Reprinted: Gen. Relativ. Gravit. 34, 1141 (2002).
- Penrose, R. and Floyd, R. M. (1971). Extraction of rotational energy from a black hole. *Nature Physical Science*, 229:177–179.
- Pryor, C. and Meylan, G. (1993). Velocity dispersions for Galactic globular clusters. In Djorgovski, S. G. and Meylan, G., editors, *Structure and Dynamics of Globular Clusters*, volume 50 of *ASP Conference Series*, page 357. Astronomical Society of the Pacific.
- Rauch, K. P. and Tremaine, S. (1996). Resonant relaxation in stellar systems. *New Astronomy*, 1:149–170.
- Reynolds, C. S. (2021). Observational constraints on black hole spin. *Annual Review of Astronomy and Astrophysics*, 59:117–154.

- Sandberg, A. (1999). The physics of information processing superobjects: Daily life among the jupiter brains. *Journal of Evolution and Technology*, 5:1–34.
- Sandberg, A., Armstrong, S., and Ćirković, M. M. (2016). That is not dead which can eternal lie: the aestivation hypothesis for resolving Fermi’s paradox. *Journal of the British Interplanetary Society*, 69(11):406–415.
- Schwinger, J. (1951). On gauge invariance and vacuum polarization. *Physical Review*, 82:664–679.
- Scoggins, M. T. and Kipping, D. (2023). Lazarus stars: numerical investigations of stellar evolution with star-lifting as a life extension strategy. *Monthly Notices of the Royal Astronomical Society*, 523:3251–3257.
- Shapiro, S. L. and Teukolsky, S. A. (1983). *Black Holes, White Dwarfs, and Neutron Stars: The Physics of Compact Objects*. Wiley-Interscience, New York.
- Smart, J. M. (2012). The transcension hypothesis: Sufficiently advanced civilizations invariably leave our universe, and implications for METI and SETI. *Acta Astronautica*, 78:55–68.
- Socas-Navarro, H., Haqq-Misra, J., Wright, J. T., Kopparapu, R., Benford, J., Davis, R., and TechnoClimes 2020 workshop participants (2021). Concepts for future missions to search for technosignatures. *Acta Astronautica*, 182:446–453.
- Soltis, J., Casertano, S., and Riess, A. G. (2021). The parallax of ω Centauri measured from Gaia EDR3 and a direct, geometric calibration of the tip of the red giant branch and the Hubble constant. *The Astrophysical Journal Letters*, 908:L5.
- Souza, S., Neumayer, N., Seth, A. C., Wang, Z., Clontz, C., Häberle, M., Nitschai, M. S., Smith, P. J., Matsuno, T., et al. (2026). oMEGACat. X. shedding light on the disrupted dwarf galaxy of Omega Centauri. *arXiv e-prints*, page arXiv:2603.23589. Submitted to ApJ.
- Strokov, V., Fragione, G., Wong, K. W. K., Helfer, T., and Berti, E. (2022). Hunting for intermediate-mass black holes with LISA binary radial velocity measurements. *Physical Review D*, 105:124048.
- Swanson, T. (2026a). Engineered intermediate-mass black hole systems: Infrastructure constraints, observable residue, and a multi-messenger adjudication framework. Preprint; omegacentauri.me.
- Swanson, T. (2026b). A joint radio–infrared–x-ray bound on bondi-fed accretion onto the candidate intermediate-mass black hole in omega centauri. Preprint; omegacentauri.me.
- Tarter, J. (2001). The search for extraterrestrial intelligence (SETI). *Annual Review of Astronomy and Astrophysics*, 39:511–548.
- Tchekhovskoy, A. (2015). Launching of active galactic nuclei jets. In Contopoulos, I., Gabuzda, D., and Kylafis, N., editors, *The Formation and Disruption of Black Hole Jets*, volume 414 of *Astrophysics and Space Science Library*, pages 45–82. Springer, Cham.
- Tchekhovskoy, A., Narayan, R., and McKinney, J. C. (2011). Efficient generation of jets from magnetically arrested accretion on a rapidly spinning black hole. *Monthly Notices of the Royal Astronomical Society: Letters*, 418:L79–L83.

- The KM3NeT Collaboration (2025). Observation of an ultra-high-energy cosmic neutrino with KM3NeT. *Nature*, 638(8050):376–382.
- Thorne, K. S. (1974). Disk-accretion onto a black hole. II. Evolution of the hole. *The Astrophysical Journal*, 191:507–519.
- Thorne, K. S. (1994). *Black Holes and Time Warps: Einstein’s Outrageous Legacy*. W. W. Norton, New York.
- Thorne, K. S., Price, R. H., and Macdonald, D. A. (1986). *Black Holes: The Membrane Paradigm*. Yale University Press, New Haven.
- Tipler, F. J. (1980). Extraterrestrial intelligent beings do not exist. *Quarterly Journal of the Royal Astronomical Society*, 21:267–281.
- Toffoli, T. (1980). Reversible computing. In de Bakker, J. and van Leeuwen, J., editors, *Automata, Languages and Programming (ICALP 1980)*, volume 85 of *Lecture Notes in Computer Science*, pages 632–644, Berlin, Heidelberg. Springer.
- Tremou, E., Strader, J., Chomiuk, L., Shishkovsky, L., Maccarone, T. J., Miller-Jones, J. C. A., et al. (2018). The MAVERIC survey: Still no evidence for accreting intermediate-mass black holes in globular clusters. *The Astrophysical Journal*, 862:16.
- van der Marel, R. P. and Anderson, J. (2010). New limits on an intermediate-mass black hole in Omega Centauri. II. Dynamical models. *The Astrophysical Journal*, 710:1063–1088.
- Vidal, C. (2011). Black holes: Attractors for intelligence? arXiv:1104.4362; presented at the Kavli Royal Society International Centre, “Towards a scientific and societal agenda on extra-terrestrial life”.
- Vidal, C. (2014). *The Beginning and the End: The Meaning of Life in a Cosmological Perspective*. The Frontiers Collection. Springer, Cham.
- Webb, S. (2015). *If the Universe Is Teeming with Aliens. . . Where Is Everybody? Seventy-Five Solutions to the Fermi Paradox and the Problem of Extraterrestrial Life*. Science and Fiction. Springer, Cham, 2 edition.
- Whitaker, M., Kerr, E., Seth, A., Häberle, M., Strader, J., Anderson, J., Bellini, A., Clontz, C., Freeman, Z., Griggio, M., Kamann, S., Libralato, M., Neumayer, N., González Prieto, E., Rodriguez, C. L., Saracino, S., Smith, P., van de Ven, G., and Wang, Z. (2026). A long period stellar-mass black hole binary in ω Centauri. *The Astrophysical Journal Letters*.
- Wolpert, D. H. (2019). The stochastic thermodynamics of computation. *Journal of Physics A: Mathematical and Theoretical*, 52:193001.
- Wright, J. T., Haqq-Misra, J., Frank, A., Kopparapu, R., Lingam, M., and Sheikh, S. Z. (2022). The case for technosignatures: Why they may be abundant, long-lived, highly detectable, and unambiguous. *The Astrophysical Journal Letters*, 927(2):L30.
- Wright, J. T., Mullan, B., Sigurdsson, S., and Povich, M. S. (2014). The $\hat{g}$ infrared search for extraterrestrial civilizations with large energy supplies. I. Background and justification. *The Astrophysical Journal*, 792(1):26.

Zocchi, A., Gieles, M., and Hénault-Brunet, V. (2019). The effect of stellar-mass black holes on the central kinematics of ω Cen: a cautionary tale for IMBH interpretations. *Monthly Notices of the Royal Astronomical Society*, 482:4713–4725.

A Staged exploitation architecture

Epistemic status: This section is speculative civilizational engineering. It is included not as prediction but because the falsification program of Section 6 requires a concrete model of what would be observable at each stage. Each stage uses only physics established in Section 4 plus engineering extrapolations that are individually defended in the cited literature; the conjunction is unargued-for and carries the usual burden of compounded speculation.

We model the migration as five phases (Fig. 8), parameterized for a civilization departing a planetary system ~ 5 kpc from ω Cen.

Phase 1 — Scouts. Gram-scale beamed-sail probes at $0.1\text{--}0.2\,c$ (Lubin, 2016), with photon-braking and magnetic-sail deceleration into the cluster halo; transit $\sim 10^5$ yr. Objectives: resolve the IMBH-versus-swarm question in situ, map the rocky-body inventory, and emplace a deceleration relay for following waves. *External signature: none detectable at interstellar range.*

Phase 2 — Self-replicating industry. Kilogram-scale seed factories bootstrap exponentially from asteroidal material, following the NASA self-replicating-systems concept (Freitas and Gilbreath, 1982); decades to centuries to 10^{3+} factory units and a relay laser. *Signature: waste heat of order early-industrial — far below detectability.*

Phase 3 — Compute swarm. The main payload (substrate-independent minds, by this stage the default cargo; cf. Smart 2012) arrives and assembles a free-flying compute swarm. While the hole is being fed, the swarm’s inner working boundary sits at $\gtrsim 10^2\,r_g$ ($\sim 3 \times 10^6$ km at $2 \times 10^4\,M_\odot$): the companion infrastructure analysis (Swanson, 2026a) finds that accretion-flow erosion excludes hardware inside $\sim 30\,r_g$ of a fueled hole, and differential Lense–Thirring precession already excludes rigid formations at $10^2\,r_g$, so the swarm must be free-flying there in any case. Controlled star-lifting of brown dwarfs establishes a thin sub-Eddington disk; BZ extraction begins at low spin. Tiered orbits allocate clock rate (Section 4.4), though for an operating (fueled) system the stable dilation on offer at the $10^2\,r_g$ working boundary is modest, $d\tau/dt \simeq 0.985$; the deep-orbit clock rates of Fig. 6 become available only to dormant or archival elements around an unfueled hole. *Signature: faint, soft accretion luminosity — the one phase in which the system would be conventionally observable; at ω Cen, current limits already exclude an active Phase 3 fed above $\sim 4 \times 10^{-3}$ of the Bondi rate (Mahida et al., 2026) or radiating above $\sim 10^{-12}$ of Eddington (Haggard et al., 2013; Chen et al., 2026).*

Phase 4 — Spin-up. Sustained prograde feeding ($\sim 10^{4-5}$ stars over $3.4\text{--}5.3 \times 10^7$ yr, 0.79 Salpeter e-foldings to reach Thorne’s radiation-limited $a_\star = 0.998$ ($M_f/M_i \approx 2.2$) (Bardeen, 1970; Thorne, 1974; Swanson, 2026a); Section 4.1) drives $a_\star \rightarrow 0.9\text{--}0.998$; the ISCO migrates inward by a factor ~ 5 and jet efficiency crosses unity in MAD episodes; the fueled swarm holds its $\sim 10^2\,r_g$ working boundary throughout (Swanson, 2026a). *Signature: episodic jet activity if imperfectly managed; high spin as a permanent record. Secular cluster-core depletion is a weaker observable than earlier drafts implied: the accreted $2.4 \times 10^4\,M_\odot$ (Section 4.1) is 22.5 per cent of the stellar mass inside 2 pc but only 1.29 per cent of the mass inside the 10-pc reservoir the fuel census requires, recomputed directly against the enclosed-mass profile of Section 5.2 (Table 5), the removed stellar mass reappears as central point mass so the total enclosed mass profile is preserved. The stellar mass function is not, and that is the observable P4c names. The refill clock was previously quoted as the ~ 4 -Gyr core relaxation time; that is the right clock for the 2-pc deficit ($\log t_{rc} = 9.60$) and the wrong one for the 10-pc reservoir the fuel census actually requires, whose half-mass relaxation time is $\log t_{rh} = 10.09$, or ~ 12 Gyr (Harris, 1996). The 10-pc deficit is therefore more durable than earlier drafts allowed, on the order of the cluster’s age rather than a third of it, and the correction runs against the hedge we had drawn from it (Swanson, 2026a). We flag the strongest hidden premise of the whole architecture here: Phases 3–5 presuppose institutional or goal stability over 10^8 yr, longer than mammalian evolutionary history, and stellar feeding requires active*

orbit-shaping of individual stars with non-trivial Δv budgets. We know of no principled argument that either is achievable; the architecture is offered conditional on both.

The Δv can at least be priced to order of magnitude, and the pricing rules out two of the three options a reader might imagine. Delivering a star inward from a near-circular cluster orbit requires removing nearly all of its orbital angular momentum; the per-star impulse is of order the local orbital speed, tens of km s^{-1} across the central 1–10 pc and growing with the radius of the ~ 10 -pc reservoir that the fuel census of Section 3.1 requires. The energy is not the obstacle (binding energy is released on the way down); the angular momentum is.

Natural refilling does not supply the rate. Phase 4 needs 10^{4-5} stars over $3.4\text{--}5.3 \times 10^7$ yr, i.e. $\sim 1\text{--}2 \times 10^{-3}$ stars yr^{-1} for the 5.6×10^4 stars the mass budget implies. Natural loss-cone refilling, which resonant relaxation drives near the influence radius (Rauch and Tremaine, 1996), delivers stars to the hole at the tidal-disruption rate, $\sim 10^{-8}\text{--}10^{-7}$ yr^{-1} for IMBHs in evolved clusters and $\sim 5 \times 10^{-8}$ yr^{-1} in the ω Cen models of González Prieto et al. (2025) specifically (Swanson, 2026a). The shortfall is four to five orders of magnitude, and the repaired episode duration of Section 4.1 moves it by less than a factor of two. The natural channel is not a slower alternative to engineered delivery; it is not a fuel supply for this architecture at all, and essentially the whole Phase-4 budget must be actuated.

Beamed-momentum tugs are excluded, on the cluster’s own light rather than on the fuel ceiling. The kinetic work is not the problem. Imparting Δv to the $2.4 \times 10^4 M_\odot$ that Phase 4 consumes costs $\frac{1}{2} M \Delta v^2$, which averaged over the 5.3×10^7 yr episode of Section 4.1 is $15 L_\odot$ at 20 km s^{-1} and $3.7 \times 10^2 L_\odot$ at 10^2 km s^{-1} . Paying the same momentum in photons costs $E = \Delta p c$, larger by $2c/\Delta v$, which is a factor of 3×10^4 at 20 km s^{-1} .

Earlier drafts priced that against a ceiling of $7.6 \times 10^5 L_\odot$ described as “the JWST-limited accretion luminosity”. Two corrections are needed. First the label: $7.6 \times 10^5 L_\odot$ is the ambient Bondi accretion power at the fiducial mass and spin, taken from the companion infrastructure analysis (Swanson, 2026a); the JWST limit of test T1 is 2.5×10^{26} W, which is $0.65 L_\odot$, and the two differ by a factor of 10^6 because they are different quantities. Second the verdict. Against the Bondi ceiling the exclusion is not robust: at the 10^2 km s^{-1} earlier drafts assumed, the photon tug costs $2.2 \times 10^6 L_\odot$, or $2.9\times$ the ceiling, but the impulse this section itself calls for is of order the local orbital speed, and the King profile of Section 5.2 gives 8.2, 15.2 and 28.3 km s^{-1} at 1, 2 and 10 pc. At 20 km s^{-1} the photon tug costs $4.5 \times 10^5 L_\odot$, which is 0.59 of the ceiling and therefore fits; the exclusion threshold sits at $\Delta v \gtrsim 34 \text{ km s}^{-1}$, above the whole orbital-speed bracket.

The exclusion survives on a comparator that does not depend on the contested ceiling. The photons are absorbed by the star being moved and reradiated at stellar temperature, so they are added to the cluster’s integrated light, and ω Cen radiates $1.09 \times 10^6 L_\odot$ in total at $M_{V,I} = -10.26$. The photon tug costs 17 per cent of that at 8.2 km s^{-1} , 41 per cent at 20, and 58 per cent at 28.3: a diffuse excess that ordinary photometry of the cluster excludes at every Δv in the bracket. The kinetic channel over the same bracket costs between 2×10^{-6} and 3×10^{-5} of the cluster’s light. Gravitational-assist sequencing among cluster stars, which uses other stars as momentum reservoirs and pays only actuator overhead, is the one scheme that survives its own thermal signature; the conclusion is the one earlier drafts reached, and the margin behind it is now three to four orders of magnitude rather than a factor of a few.

Delivery geometry, not merely delivery rate, is what the spin fossil records. Loss-cone stars arrive isotropically, and isotropic feeding random-walks a hole’s spin rather than raising it (Section 6; Swanson 2026a). Coherent prograde delivery is therefore not an efficiency improvement on the natural channel but the only channel that produces the observable this paper treats as its decisive test. A full account (which stars, in what order, at what actuator cost, in what orbital orientation) has not been produced and remains the largest quantitative gap in this section.

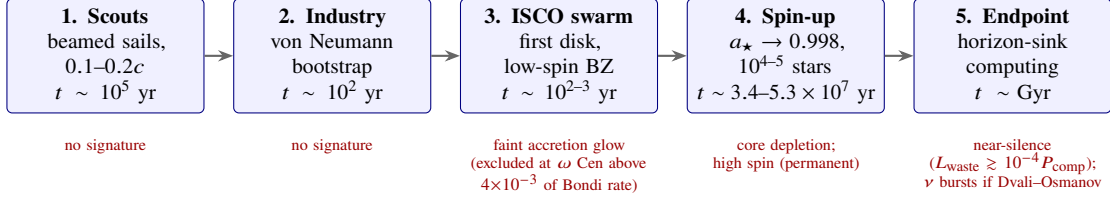


Figure 8: The five-phase exploitation architecture (speculative; see epistemic-status box) and the external signature of each phase. Only Phase 3 is conventionally luminous; the permanent observables are dynamical (spin, core depletion), which is why gravitational-wave spin measurement is the decisive test.

Phase 5 — Endpoint. Near-extremal spin; waste entropy disposed across the horizon (Section 4.2); residual stellar population dispersed or consumed. During fueled operation the swarm computes at the $\gtrsim 10^2 r_g$ working boundary ($d\tau/dt \simeq 0.985$); if the system instead enters a dormant, unfueled archival mode, elements can descend to the near-ISCO orbits of Fig. 6 and bank the full $\sim 11:1$ stable dilation. The system is nearly, but not perfectly, thermodynamically silent: transport losses in delivering waste entropy to the horizon impose a waste-heat floor $L_{\text{waste}} \gtrsim 10^{-4} P_{\text{comp}}$ (Swanson, 2026a), so current JWST mid-infrared limits already bound the computational throughput of any operating Phase-5 system at ω Cen to $P_{\text{comp}} \lesssim 10^{3-4} L_{\odot}$; there is no radio leakage, and the horizon temperature is in the picokelvin range (Eq. 4). The sole conjectured emission channel is burst-mode: if intractable computations are delegated to manufactured micro black holes per Dvali and Osmanov (2023), their evaporation yields high-energy neutrino and gamma-ray transients. We note for completeness that the formation of micro black holes from focused radiation may be blocked by Schwinger-limit pair production (Álvarez-Domínguez et al., 2024; Breit and Wheeler, 1934; Schwinger, 1951); this conclusion is contested (Loeb, 2024), and the MTH does not depend on the channel — it affects only the strength of prediction T6 below.